

Corner polyhedra and Schnyder labelings: polynomial exponents and limiting amplitudes

OEIS A377922, A377920 and A377921

Part I: research note; Part II: research addendum; Part III: research report 112
merged in ProveIt from two manuscripts (batch 77, sources 45 and 46)
and bundle Report 112 (batch 103, source 47; Part III)

2 October 2026; Part III added 5 October 2026

Abstract

Fusy, Narmanli and Schaeffer encode corner polyhedra (A377922) and 3-connected Schnyder labelings (A377920) as quadrant walks whose steps depend on a two-state parity, and conjectured equivalents $\kappa\mu^n n^{-\alpha}$ with irrational exponents. Part I proves two-sided Θ bounds with the conjectured exponents, non-D-finiteness of the generating functions of A377922, A377920 and A377921, and first-threshold inverses with $O(1)$ error. Part II proves the full equivalents with finite positive amplitudes, including $\tilde{s}_n \sim (16/19)^3 \kappa_S \mu_S^n n^{-\alpha_S}$ for A377921, and a Lambert- W inverse with a vanishing-width ceiling bracket. No digits of the amplitudes are certified. The two manuscripts are printed in full; a guide fixes the notation shared between them.

[5 October 2026, batch 103.] Part III adds bundle Report 112, which proves the exponents of A377922, A377920 and A377921 again, at the logarithmic scale, as instances of a general cone theorem for walks with a finite internal state and unbounded steps (Theorem 24.1). New with it: that theorem, a six-face sleeve injection giving $s_{n-6} \leq \tilde{s}_n \leq s_n$, a non-D-finiteness criterion that needs only the logarithmic law, a fixed-endpoint sandwich for A377922 and an explicit exponential-moment certificate.

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Guide to the merged report

What the report proves

Let p_n count corner polyhedra with $n + 3$ flats (A377922), s_n count 3-connected Schnyder labelings with n inner faces (A377920), and $\tilde{s}_n$ count rigid orthogonal surfaces (A377921), all indexed as in Fusy–Narmanli–Schaeffer [1]. Put $\mu_P = 9/2$, $\mu_S = 16/3$, $\alpha_P = 1 + \pi/\arccos(9/16)$ and $\alpha_S = 1 + \pi/\arccos(22/27)$.

- **Part I** (source 45): $p_n = \Theta(\mu_P^n n^{-\alpha_P})$ and $s_n = \Theta(\mu_S^n n^{-\alpha_S})$ on the full integer sequence (Theorem 1.1), which proves the exponents of [1, Conjecture 25]; the exponents are irrational (Lemma 9.1); the generating functions of A377922, A377920 and A377921 are not D-finite (Corollary 9.4); and the first threshold index satisfies $N(Y) = (\log Y + \alpha \log \log Y)/\log \mu + O(1)$ (Lemma 9.5).
- **Part II** (source 46): finite positive κ_P, κ_S with $p_n \sim \kappa_P \mu_P^n n^{-\alpha_P}$, $s_n \sim \kappa_S \mu_S^n n^{-\alpha_S}$ and $\tilde{s}_n \sim (16/19)^3 \kappa_S \mu_S^n n^{-\alpha_S}$ (Theorem 11.1): the two positive-amplitude equivalents of [1, Conjecture 25] and a coefficientwise equivalent for A377921. On the way: a boundary-shift comparison (Lemma 14.1), harmonic and endpoint limits for valid regeneration cycles (Theorem 15.1), exact counted-time survival constants (Theorem 16.1), a uniform killed local limit (Lemma 17.1), fixed-endpoint bridge amplitudes (Theorem 18.1) with the universal Brownian factor evaluated (Section 18.1), and a Lambert- W_{-1} inverse with a vanishing-width two-ceiling bracket (Theorem 20.1).

Part II supersedes none of Part I: it uses Part I’s models, covariance calculations, cycle buffer, survival bound, free local theorem and uniform endpoint estimate as inputs, and it does not reprove non-D-finiteness (which needs only the Θ bounds and the irrationality of the exponents).

[**write, 2026-10-05**] **Part III** (source 47, bundle Report 112, batch 103) proves $p_n = \mu_P^n n^{-\alpha_P + o(1)}$, $s_n = \mu_S^n n^{-\alpha_S + o(1)}$ and the same for $\tilde{s}_n$, non-D-finiteness and inverses with error $o(\log \log Y)$ (Theorem 23.1): a second route to Part I, weaker than Parts I–II. New to the report: a logarithmic cone theorem for Markov-additive walks with a finite internal state, unbounded steps and a uniform exponential moment (Theorem 24.1); the sleeve injection $s_{n-6} \leq \tilde{s}_n \leq s_n$ (Proposition 29.1), which with Theorem 1.1 gives $\tilde{s}_n = \Theta(\mu_S^n n^{-\alpha_S})$ (Remark 29.3); a non-D-finiteness criterion that needs only the logarithmic law (Remark 27.1); the sandwich $e_{n-1} \leq p_n \leq e_{n+1}$; the moment certificate 189/13; the bounds $4 < \alpha_P < 5$, $6 < \alpha_S < 7$; and three references. Section 22 maps every result of Report 112 to Parts I–II.

Sources and provenance

Both manuscripts are dated 2 October 2026 and name no ProveIt commit; neither inspected the repository beyond a GitHub code search (which then returned nothing for these sequences). They arrived in batch 77 of docs/incoming (arrival commit 096ee7b87) and were placed in aa7345800. Author lines: “Research note” (source 45) and “Research addendum” (source 46); the PDF metadata of both say “Research note”. Neither names a tool; the report is unrefereed and not formalized.

Source	Archive (wrapper directory, main file)	Pin	Placed	Printed as
45	oeis-bimodal-cone-research.zip (oeis-bimodal-cone-report/ article.tex, 15-page PDF)	none	aa7345800	Part I: Sections 1–10, Appendix A
46	oeis-bimodal-full-asymptotics.zip (oeis-bimodal-amplitude-report/ addendum.tex, 14-page PDF)	none	aa7345800	Part II: Sections 11–21

[write, 2026-10-05] Source 47 is bundle Report 112, `Geometric_OEIS_Sequences_Cone_Exponents_and_Inverses_Source.zip` (wrapper `report112/`, main file `report112.tex`, 19-page PDF), dated 2 October 2026, author line “Research report 112”, no pin; arrival commit `60f54ea06`, placed in `9c995cefe` (batch 103), printed as Part III, Sections 22–32. It was written without knowledge of sources 45 and 46. Section 22.2 gives the details.

How the two sources were merged

- Source 46 is an addendum to source 45 and embeds an unchanged copy of all 30 files of source 45’s package (verified byte-identical at placement). That copy is not shipped; source 45 is printed once, as Part I.
- Part I keeps source 45’s numbering exactly (Sections 1–10, Appendix A), because this guide is unnumbered. So the numbers in source 46’s “Foundation Lemma 2.1, Lemma 4.1, Section 3, Section 4.1, Proposition 6.1, Proposition 7.2, equation (8.6)” are still correct here; they are now hyperlinks. Source 46’s Section n is Section $n + 10$ here, and its Theorem $n.m$ is Theorem $(n + 10).m$.
- Both manuscripts are printed in full, with their abstracts. Source 46’s Section 2 (Section 12 here) restates Part I’s inputs in its own notation; it is kept, because the rest of Part II cites its displays. No result is proved in both Parts with the same strength: Part II’s inverse theorem sharpens Part I’s inverse lemma, and both are printed.
- Source 45’s appendix of exact checks is placed after Part II, so that it stays Appendix A.
- Labels carry the prefix `cps:` in Part I and `cps:amp:` in Part II. Apart from labels, the only changes to the manuscripts’ text are: source 46’s citations of the Foundation, which became cross-references to Part I; its citation key `DW`, which became source 45’s `DWrevisited` (the same paper); and the dated notes marked [write]. No statement, proof, symbol or number was changed.
- One bibliography: source 45’s, plus source 46’s entry for Bañuelos–Smits. Source 46’s entries for Fusy–Narmanli–Schaeffer and Denisov–Wachtel are the same papers as source 45’s. Its `Foundation` entry is replaced by Part I. Its uncited `OEIS` entry (`A377922`, `A377920`, `A377921`) duplicates source 45’s two `OEIS` entries and is omitted. Source 45’s uncited entry for Denisov–Wachtel 2015 is kept, as delivered.

[write, 2026-10-05] Part III was added after Part II and before Appendix A, which keeps its letter. Its labels carry the prefix `cps:log:`; its Section 22 is a guide written then, and Report 112’s Section k is Section $k + 22$. No earlier number or label changed. Section 22.3 lists how Report 112 is printed and how its bibliography was merged.

Notation shared and not shared

The two manuscripts use the same normalizations: $\mu_P, \mu_S, \alpha_P, \alpha_S$, the lattice L , the kernels K_P, K_S , the killed transition probabilities q_n and q_n^* , the covariances Σ_P, Σ_S per counted step (Part I remarks that the matrices of [1, Section 4.2] are $\mu\Sigma$), the angle θ and $p = \pi/\theta = \alpha - 1$. *No symbol was renamed and no normalization was changed*; each Part keeps its own letters, and the letters below mean different things in the two Parts, or in different places of one Part. In both Parts the roman e is the even parity state, not Euler’s number.

[write, 2026-10-05] Part III has its own table, Table 3. The most dangerous clash: Report 112 works in half-scale coordinates ($x = 2U + c$), so its Σ_P, Σ_S and phase covariances are one quarter

Table 1: Clashing letters. “Reading to avoid” is the tempting false identification.

Letter	Part I (source 45)	Part II (source 46)	Reading to avoid
D	$D = (1 - \frac{1}{3x^2})(1 - \frac{y^2}{3})$ in (2.4); D_0 ; operators $D_x = x\partial_x$, D_y	D : valid complete-cycle kernel (Section 13), D_i per parity; Part I’s D is D_P	D^m in Part II is not a power of Part I’s denominator
H	$H = 1 - x^{-2} - y^2 - 2x^{-2}y^2$ in (2.11)	$H_S =$ Part I’s H ; $\mathcal{H} = \lim D^m V$ harmonic function	$\mathcal{H}$ is not H
T	return times T_k ; $T(z) = \sum \tilde{s}_n z^n$ (proof of Corollary 9.4)	whitening $T = (2\Sigma)^{-1/2}$; T_k ; $T_Y = \log Y$; Part I’s $T(z)$ is $\tilde{S}(z)$	T in Part II is a matrix, not a series
K	tilted kernels K_P, K_S	the same kernels; also the wedge $K = TC$, K_b , heat kernels k_t^K , the semigroup K_t	
C	generic constants; conditional covariances C_i ; a fixed ball C (Section 8)	open quadrant $C = (0, \infty)^2$; C_B ; bridge amplitude $\mathcal{C}(a, b)$; shifts c_P, c_S	
B	entry $B = x/y + x^{-1}y/H$; balls B, B', B_n	continuity sets B ; bridge factor B_Σ ; shifts b_P, b_S ; coefficients b_j	
L	lattice L ; $L = \log Y$ in the proof of Lemma 9.5	lattice L ; $L_m = \sqrt{m} \log m$	
A	generating function $A(z)$ (Lemma 9.2)	survival amplitudes $A(a)$, $A^*(b)$	
S	survival $S_n(a)$; $S(z) = \sum s_n z^n$	$S(z)$ the same series; S also names the model	
β	$\beta = k - \alpha$ (proof of Lemma 9.2)	survival constants β, β^*, β_B	
λ	principal eigenvalue $\lambda(t)$; exponential-moment parameter	$\lambda = \log \mu$ (Theorem 20.1)	
h	corrector h_e, h_o	corrector; also $h(x) = u(\Sigma^{-1/2}x)$ (Section 18.1)	
a	stay probability $a = 1/6$ or $1/8$ in (3.1); also points	points; sequences a_n in Theorem 20.1	
r, g	coordinate index r ; $r \in \{9/16, 22/27\}$	$r(Y)$ the Lambert centre; g_t entrance density, $g(t)$ in a proof	
γ, κ	—	κ_P, κ_S amplitudes; γ a generic amplitude ($= \kappa$)	

of the matrices with the same names in Parts I–II (Remark 22.1); the angles and exponents agree. Its Γ_P, Γ_S are μ_P, μ_S and its ν is p .

What is not claimed

Every limitation stated by either source is kept in its text. In summary:

- No numerical digits of κ_P or κ_S , no rate of convergence and no correction term are certified (Part II, Sections 11 and 21). The amplitudes are characterized by positive killed-harmonic and Brownian heat-kernel expressions; only the universal Brownian factor is evaluated.
- No unconditional rounding formula $N(Y) = \lceil r(Y) \rceil$ (Part II); Part I’s inverse has $O(1)$ error and no next constant.
- A377923, whose OEIS entry the delivery also exported, is a different sequence and is outside

both theorems.

- No general cone theorem for Markov-modulated walks (Part I, Section 1); the arguments use a fixed cycle buffer specific to these two models. No cone-killed local theorem for the two-state chain is assumed (Part II).
- The counting bijections are Fusy–Narmanli–Schaeffer’s, the iid cone input is Denisov–Wachtel’s, and the Brownian cone kernel is Bañuelos–Smits’. Neither source claims publication priority; both literature checks (`45-cone-literature-status.md`, `46-amp-literature-status.md`) are bounded snapshots that do not establish novelty.
- The finite computations check identities, normalizations and indexing (coefficients to $n = 30$); they do not establish any limit. Neither source was externally peer reviewed or machine-formalized; the shipped mathematical reviews are the deliveries’ own.

[**write, 2026-10-05**] The bullet “No general cone theorem for Markov-modulated walks” is true of Parts I–II, but no longer of the report: Part III’s Theorem 24.1 is a general theorem for walks with a finite internal state, at the logarithmic scale. It now holds only for amplitudes and Θ bounds: the report has no general amplitude or Θ theorem (Questions Q1 and Q2). Part III keeps all of Report 112’s non-claims: no amplitude, no full equivalent, no rate, no claim at endpoints without seeds, no $O(1)$ inverse of its own, and finite checks that prove no limit.

Inversion mechanics and formal status

Part I’s Lemma 9.5 and Part II’s Theorem 20.1 are instances of three theorems of the repository’s transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`): `p0:thm:lambert-core`, `p0:thm:lambert-centered` and, for the integer bracket, `p0:thm:staircase`. No novelty is claimed for the inversion mechanics. Nothing in this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The only related formal statement is the generic staircase lemma `Fabius.staircase_separation` (`Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`), which concerns an arbitrary monotone interpolation, not these sequences.

[**write, 2026-10-05**] Part III’s inverse (23.4) is the same instance of `p0:thm:lambert-core` at logarithmic precision. Nothing in Part III is formalized either.

Part I

Polynomial exponents (source 45)

Polynomial exponents for corner polyhedra and Schnyder labelings

Research note, 2 October 2026

Abstract of source 45, as delivered. The quadrant-walk models of Fusy, Narmanli, and Schaeffer have a two-state parity dependence that prevents a direct application of an independent-increment cone theorem at the original time scale. We prove two-sided constant-factor estimates for the two enumerating sequences:

$$p_n = \Theta\left((9/2)^n n^{-1-\pi/\arccos(9/16)}\right), \quad s_n = \Theta\left((16/3)^n n^{-1-\pi/\arccos(22/27)}\right).$$

The proof regenerates at returns to a parity state. A deterministic one-unit bound on excursions within a regeneration cycle transfers independent-increment cone estimates to the original walk. Exponential concentration of the random clock yields lower bounds at every sufficiently large exact time; a Fourier local limit estimate and an interior gluing argument yield the extra factor n^{-1} . The estimates prove the conjectured polynomial exponents, but do not prove convergence to positive asymptotic amplitudes. They also imply non-D-finiteness of the two generating functions and, by a rational generating-function identity, of the related rigid-surface generating function. All symbolic and finite enumeration checks are supplied in a reproducibility package.

1 Results and scope

Let p_n be the number of polyhedral orientations with n inner vertices, equivalently the number of combinatorial types of corner polyhedra with $n + 3$ flats. Let s_n count 3-connected Schnyder labelings with n inner faces in the associated quadrangular dissection. These are the sequences [A377922](#) and [A377920](#), with precisely the indexing of Fusy–Narmanli–Schaeffer [1, 11, 12]. Put

$$\mu_P = \frac{9}{2}, \quad \mu_S = \frac{16}{3}, \quad \alpha_P = 1 + \frac{\pi}{\arccos(9/16)}, \quad \alpha_S = 1 + \frac{\pi}{\arccos(22/27)}. \quad (1.1)$$

Throughout, $a_n = \Theta(b_n)$ means that positive constants c, C exist such that $cb_n \leq a_n \leq Cb_n$ for all sufficiently large integers n .

Theorem 1.1. *With the preceding conventions,*

$$p_n = \Theta(\mu_P^n n^{-\alpha_P}), \quad s_n = \Theta(\mu_S^n n^{-\alpha_S}). \quad (1.2)$$

Both estimates hold on the full integer sequence, not merely on a congruence subsequence.

Sequence	Objects	Exponential rate	Polynomial exponent
A377922	Corner polyhedra	9/2	$\alpha_P = 4.227476082\dots$
A377920	Schnyder labelings	16/3	$\alpha_S = 6.080304846\dots$

The exponential rates were proved in [1, Theorem 24]. Their Conjecture 25 predicts asymptotic equivalents with positive amplitudes and the exponents in (1.1). Theorem 1.1 establishes the exponents and stronger two-sided bounds, but leaves the amplitude limits unresolved. In particular,

$$\lim_{n \rightarrow \infty} \frac{\log p_n - n \log \mu_P}{\log n} = -\alpha_P, \quad \lim_{n \rightarrow \infty} \frac{\log s_n - n \log \mu_S}{\log n} = -\alpha_S. \quad (1.3)$$

The manuscript relies on the published combinatorial bijections and on the independent-increment cone theorem stated in Section 5. It does not posit a new general cone theorem for arbitrary Markov-modulated walks. Its argument is specific to a geometric property of these two tandem-walk models.

[**write, 2026-10-05**] This disclaimer describes source 45 alone. Part III (bundle Report 112) proves a general cone theorem for walks with a finite internal state, Theorem 24.1: under hypotheses (H1)–(H5), which allow unbounded steps with a uniform exponential moment and need no cycle buffer, the killed fixed-endpoint kernel is $n^{-1-\pi/\theta+o(1)}$ at every integer time. It is a logarithmic-scale statement; for amplitudes and for Θ bounds the disclaimer still holds (Questions Q1–Q2).

[**write, 2026-10-02**] The amplitude limits that this paragraph leaves unresolved are established in Part II (source 46): Theorem 11.1 proves both positive-amplitude equivalents of [1, Conjecture 25], on the full integer sequence, and an equivalent for A377921. Part II characterizes κ_P, κ_S by positive harmonic and heat-kernel expressions and certifies no numerical digits of them.

2 Exact walk models and normalization

A tandem step is either $\text{SE} = (1, -1)$ or a *face step* $(-i, j)$ with $i, j \geq 0$. A walk is even when every increment has coordinates of the same parity. The common state space is

$$L = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod{2}\}. \quad (2.1)$$

The internal state J is the parity of the second coordinate, denoted e or o. In a matrix of Laurent series, rows give the initial state, columns the final state, and the coefficient of $x^a y^b$ gives the weight of displacement (a, b) . Nonnegative coefficients are understood in the indicated convergent expansions.

2.1 Corner polyhedra

The P -admissible steps are

$$\begin{aligned} \mathcal{S}_P^e &= \{(1, -1)\} \cup \{(-2i, 2j) : i \geq 0, j \geq 1\} \cup \{(-2i - 1, 2j + 1) : i, j \geq 0\}, \\ \mathcal{S}_P^o &= \{(1, -1)\} \cup \{(-2i, 2j) : i \geq 1, j \geq 0\} \cup \{(-2i - 1, 2j + 1) : i, j \geq 0\}. \end{aligned} \quad (2.2)$$

Every step has counting weight one. Proposition 13 of [1] identifies p_n with the number of P -admissible walks of length n from $(0, 0)$ to the horizontal axis, staying in the nonnegative quadrant. A nonempty such walk cannot return to $(0, 0)$: its final step would have to be horizontal from a point on the horizontal axis, whereas no such face step is permitted at even height. Therefore the endpoints are $(2i, 0)$, $i \geq 1$.

Write $D_0 = (1 - x^{-2})(1 - y^2)$. The unweighted matrix is

$$M_P(x, y) = \begin{pmatrix} y^2/D_0 & x/y + x^{-1}y/D_0 \\ x/y + x^{-1}y/D_0 & x^{-2}/D_0 \end{pmatrix}. \quad (2.3)$$

Tilting and normalizing gives

$$K_P(x, y) = \frac{2}{9} M_P(\sqrt{3}x, y/\sqrt{3}) = \begin{pmatrix} \frac{2y^2}{27D} & \frac{2x}{3y} + \frac{2y}{27xD} \\ \frac{2x}{3y} + \frac{2y}{27xD} & \frac{2}{27x^2D} \end{pmatrix}, \quad D = \left(1 - \frac{1}{3x^2}\right) \left(1 - \frac{y^2}{3}\right). \quad (2.4)$$

At $(1, 1)$ the matrix is stochastic. Each admissible displacement z has probability $\mu_P^{-1} 3^{(z_1 - z_2)/2}$. Let

$$Q_P = L \cap \mathbb{Z}_{\geq 0}^2, \quad q_n^P(a, b) = \mathbb{P}_a(X_n = b, X_t \in Q_P \text{ for } 0 \leq t \leq n). \quad (2.5)$$

The tilt telescopes along a path, giving the exact identity

$$p_n = \mu_P^n \sum_{i \geq 1} 3^{-i} q_n^P((0, 0), (2i, 0)). \quad (2.6)$$

2.2 Schnyder labelings

An S -admissible face step is one of the following, with $\ell, r \geq 0$ and weight $\binom{\ell+r}{r}$:

initial parity	face displacement
either	$(-2\ell - 2, 2r + 2)$
e	$(-2\ell - 1, 2r + 3)$
o	$(-2\ell - 3, 2r + 1)$

(2.7)

Weights count distinct decorations. Write s'_n for the subclass in which the two outer vertices labeled G are non-isolated. Proposition 14 and Remark 9 of [1] state that s'_n counts weighted quadrant walks from $(0, 2)$ to $(2, 0)$ with $n + 2$ southeast steps, and that

$$s_n = s'_n + 2s'_{n-1} + s'_{n-2} \quad (n \geq 4). \quad (2.8)$$

Each such walk begins and ends with a southeast step. Removing the last one leaves a walk from $(0, 2)$ to $(1, 1)$ with $n + 1$ southeast steps.

An *aggregate* consists of one southeast step followed by an arbitrary, possibly empty, string of face steps. The face-step matrix and southeast matrix are

$$W(x, y) = \frac{1}{1 - x^{-2} - y^2} \begin{pmatrix} x^{-2}y^2 & x^{-1}y^3 \\ x^{-3}y & x^{-2}y^2 \end{pmatrix}, \quad E(x, y) = \frac{x}{y} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (2.9)$$

Thus $M_S = E(I - W)^{-1}$. With

$$H = 1 - x^{-2} - y^2 - 2x^{-2}y^2, \quad B = x/y + x^{-1}y/H, \quad (2.10)$$

explicit inversion yields

$$M_S(x, y) = \begin{pmatrix} x^{-2}/H & B \\ B & y^2/H \end{pmatrix}, \quad K_S(x, y) = \frac{3}{16} M_S(2x, y/2). \quad (2.11)$$

The expansion of $(I - W)^{-1}$ accounts for every decorated face string with its correct multiplicity. It converges at the tilting point $(2, 1/2)$, and in a neighborhood thereof.

Lemma 2.1 (Exact aggregate boundary). *For an aggregate walk from $(0, 2)$ to $(1, 1)$, every underlying physical step stays in the nonnegative quadrant if and only if every aggregate endpoint stays in*

$$Q_S = \{(a, b) \in L : a \geq 0, b \geq 1\}. \quad (2.12)$$

Proof. If an aggregate begins at height at least one, its southeast step remains nonnegative. Subsequent face steps decrease the first coordinate and increase the second, so nonnegativity of the aggregate's final first coordinate controls the whole face string. Conversely, any aggregate except the last must end at height at least one to allow the following southeast step; the prescribed last endpoint has height one. The first coordinate is nonnegative at all endpoints whenever the physical path is valid. $\square$

Let q_N^S denote the K_S chain killed outside Q_S , with N counting aggregates. A decorated path of N aggregates with displacement (a, b) has probability $\mu_S^{-N} 2^{a-b}$. The total displacement here is $(1, -1)$; hence

$$s'_n = \frac{\mu_S^{n+1}}{4} q_{n+1}^S((0, 2), (1, 1)). \quad (2.13)$$

There is no physical-time offset in (2.6); the $n + 1$ in (2.13) is essential.

3 Correctors and regenerative covariance

In this section K denotes either K_P or K_S . Time means a physical step for P and an aggregate for S . In both cases the internal transition matrix is

$$P = K(1, 1) = \begin{pmatrix} a & 1-a \\ 1-a & a \end{pmatrix}, \quad a = \begin{cases} 1/6 & (P), \\ 1/8 & (S). \end{cases} \quad (3.1)$$

Its stationary distribution is $(1/2, 1/2)$. Define $D_x = x\partial_x$ and $D_y = y\partial_y$. For example, the conditional mean of coordinate r in state i is

$$d_{i,r} = \sum_j D_r K_{ij} \Big|_{x=y=1}. \quad (3.2)$$

The means satisfy $d_o = -d_e$. Solving $d_i + \sum_j P_{ij}(h_j - h_i) = 0$ gives a bounded corrector with $h_o = -h_e$:

$$\begin{array}{c|cc} & d_e & h_e \\ \hline P & (1/6, 1/6) & (1/10, 1/10) \\ S & (-1/8, -1/8) & (-1/14, -1/14). \end{array} \quad (3.3)$$

Consequently $M_t = X_t + h_{J_t}$ is a martingale.

The conditional covariance matrix of its increment in state i has entries

$$(C_i)_{rs} = \sum_j [D_r D_s K_{ij} + (h_{j,r} - h_{i,r}) D_s K_{ij} + (h_{j,s} - h_{i,s}) D_r K_{ij} + (h_{j,r} - h_{i,r})(h_{j,s} - h_{i,s}) K_{ij}]_{x=y=1}. \quad (3.4)$$

Evaluation gives

$$C_e^P = \begin{pmatrix} 12/5 & -9/5 \\ -9/5 & 4 \end{pmatrix}, \quad C_o^P = \begin{pmatrix} 4 & -9/5 \\ -9/5 & 12/5 \end{pmatrix}, \quad (3.5)$$

$$C_e^S = \begin{pmatrix} 6 & -88/21 \\ -88/21 & 30/7 \end{pmatrix}, \quad C_o^S = \begin{pmatrix} 30/7 & -88/21 \\ -88/21 & 6 \end{pmatrix}. \quad (3.6)$$

Averaging over the stationary internal chain gives the covariance per unit time:

$$\Sigma_P = \begin{pmatrix} 16/5 & -9/5 \\ -9/5 & 16/5 \end{pmatrix}, \quad \Sigma_S = \begin{pmatrix} 36/7 & -88/21 \\ -88/21 & 36/7 \end{pmatrix}. \quad (3.7)$$

Both are positive definite. The matrices displayed in [1, Section 4.2] are $\mu_P \Sigma_P$ and $\mu_S \Sigma_S$, respectively; the stochastic per-counted-step normalization here differs by these scalar factors. The correlation coefficients and conjectured powers are unchanged. The unequal matrices in (3.5) and (3.6) are one reason to avoid imposing a state-independent conditional covariance hypothesis on the original chain.

3.1 First returns to a parity state

Fix the initial parity i and set $T_0 = 0$ and $T_{k+1} = \inf\{t > T_k : J_t = i\}$. By the Markov property and translation invariance within a parity class, the pairs

$$(Y_k, \ell_k) = (X_{T_k} - X_{T_{k-1}}, T_k - T_{k-1}) \quad (3.8)$$

are independent and identically distributed. With $j \neq i$ their joint generating function is

$$R_i(u, x, y) = uK_{ii}(x, y) + \frac{u^2 K_{ij}(x, y) K_{ji}(x, y)}{1 - uK_{jj}(x, y)}. \quad (3.9)$$

The geometric term enumerates a first transition to j , any number of stays in j , and a last transition to i . The denominator is nonzero at $(1, 1, 1)$, and all coefficients are nonnegative. Analyticity there gives a joint exponential moment of (ℓ, Y) .

Differentiating (3.9), for either choice of i , yields

$$\mathbb{E}\ell = 2, \quad \mathbb{E}Y = 0, \quad \text{Cov}(Y) = 2\Sigma, \quad \text{Var}(\ell) = \begin{cases} 2/5 & (P), \\ 2/7 & (S). \end{cases} \quad (3.10)$$

For instance, $\mathbb{E}\ell = \partial_u R_i(1, 1, 1)$ and $\mathbb{E}(Y_r Y_s) = D_r D_s R_i(1, 1, 1)$. For returns to the even state,

$$\text{Cov}(\ell, Y) = \begin{cases} (-2/5, -2/5) & (P), \\ (2/7, 2/7) & (S). \end{cases} \quad (3.11)$$

Thus cycle displacement and duration are genuinely dependent. No argument below requires their independence or conditions on an exact total cycle duration.

3.2 Time reversal and cone angle

Write $K_{ij}(z)$ for a coefficient of the step matrix. The reverse transition probabilities are

$$K_{ij}^*(z) = K_{ji}(-z). \quad (3.12)$$

For a fixed endpoint in L , the total incoming mass is the relevant column sum of P , which is one. Counting measure on L is therefore invariant, and K^* is stochastic. For either killed domain,

$$q_n(a, b) = q_n^*(b, a). \quad (3.13)$$

The matrices K_P and K_S are symmetric as matrices of Laurent series, so $K^*(x, y) = K(x^{-1}, y^{-1})$. The reverse corrector is $h^* = -h$, and its corrected conditional covariance matrices are the same as (3.5)–(3.6). Reversing a first-return path preserves its probability and duration and negates its displacement. The reverse cycle law consequently has the same covariance as the forward one.

Whitening covariance 2Σ transforms the positive quadrant to a wedge with angle

$$\theta = \arccos\left(-\frac{\Sigma_{12}}{\sqrt{\Sigma_{11}\Sigma_{22}}}\right), \quad p = \frac{\pi}{\theta}, \quad \theta_P = \arccos(9/16), \quad \theta_S = \arccos(22/27). \quad (3.14)$$

Thus $\alpha = 1 + p$ in each model. Multiplying the covariance by a scalar does not change this angle.

4 A fixed buffer for regeneration cycles

Call an increment northwest if its first coordinate is nonpositive and its second coordinate is nonnegative. For P , every parity-preserving step is northwest, and every parity-changing step is either northwest or SE. The same assertion holds for S aggregates: any nonempty face string following the initial southeast step offsets its increase of the first coordinate and decrease of the second, since each face step has first coordinate at most -1 and second coordinate at least 1 . An empty aggregate is exactly SE, and cannot preserve parity.

Lemma 4.1 (One-unit cycle bound). *For a first return to either parity in either model, with endpoints a, b , every intermediate position z at the model's time scale satisfies*

$$z_r \geq \min(a_r, b_r) - 1 \quad (r = 1, 2). \quad (4.1)$$

The same statement holds for reversed cycles.

Proof. A one-step return is northwest. Otherwise the cycle consists of one parity-changing step, a string of parity-preserving northwest steps, and one parity-changing step. The first and last steps may each be northwest or SE. A drop of the second coordinate below the smaller endpoint height is possible only at the first step, when that step is SE, and its size is one. A drop of the first coordinate below the smaller endpoint first coordinate is possible only immediately before a final SE, also by one. All intervening changes have the northwest monotonicity. Reversal preserves the set of intermediate positions and exchanges the endpoints, so it preserves the bound. $\square$

If all even-parity cycle endpoints have both coordinates at least two, every intervening position has both coordinates at least one. This ensures survival in Q_P and Q_S . In the S model Lemma 2.1 then ensures survival at every underlying physical step as well. The buffer is fixed, irrespective of cycle duration or displacement.

4.1 Accessibility and finite entrance paths

The even-return cycle laws have explicit increments that make the required cone starting points accessible. In P , $(0, 2)$ is a one-step even return, and $(2, -2)$ is realized by two southeast steps. A cycle with displacement $(-2, 0)$ consists of

$$(-1, 1), \quad (-2, 0), \quad (1, -1), \quad (4.2)$$

with states e, o, o, e. The reverse cycle law therefore contains $(2, 0)$ and $(-2, 2)$. By raising the second coordinate before adding $(2, -2)$, the forward skeleton reaches arbitrarily deep diagonal points without exiting the translated cone. The analogous reverse construction first raises the first coordinate and then uses $(-2, 2)$.

In S , two empty aggregates give $(2, -2)$. A cycle of aggregate displacements

$$(1, -1), \quad (0, 2), \quad (-1, 1) \quad (4.3)$$

gives $(0, 2)$. The middle aggregate is a southeast step followed by the face step $(-1, 3)$; the last is a southeast step followed by $(-2, 2)$. A single aggregate consisting of a southeast step and $(-3, 1)$ gives the even return $(-2, 0)$. Hence the same forward and reverse accessibility argument applies.

Finally, there are positive-probability entrance paths to $(2, 2)$. For P they are

$$(0, 0) \longrightarrow (0, 4) \longrightarrow (1, 3) \longrightarrow (2, 2), \quad (4, 0) \xrightarrow{\text{reverse}} (3, 1) \longrightarrow (2, 2). \quad (4.4)$$

For S the forward aggregate path is

$$(0, 2) \xrightarrow{\text{SE}} (1, 1) \xrightarrow{\text{SE}+(-1,3)} (1, 3) \xrightarrow{\text{SE}} (2, 2). \quad (4.5)$$

Its middle aggregate passes through $(2, 0)$, which is permitted physically. The reverse entrance is

$$(1, 1) \xrightarrow{\text{reverse SE}} (0, 2) \xrightarrow{(2,0)} (2, 2). \quad (4.6)$$

The last reversed aggregate is the reversal of the valid forward path $(2, 2) \rightarrow (3, 1) \rightarrow (0, 2)$. All displayed entrances respect their killed domains.

5 Independent-increment cone input

We use the following consequences of Denisov–Wachtel [2, Theorems 2 and 3]. Let Z_k be an iid centered random walk in a planar wedge, after a nonsingular linear normalization of its covariance to the identity. Assume a nondegenerate covariance and an exponential moment. Let τ be its first exit from the open wedge and let $p = \pi/\theta$, where θ is the wedge angle. Assume $p \geq 1$, as holds for both wedges here. Then

$$\mathbb{P}_z(\tau > k) \leq C(1 + \|z\|)^p k^{-p/2}. \quad (5.1)$$

For a fixed starting point with positive killed harmonic function and any fixed ball B compactly contained in the wedge,

$$\mathbb{P}_z(\tau > k, Z_k \in \sqrt{k}B) \sim c_{z,B} k^{-p/2}, \quad c_{z,B} > 0. \quad (5.2)$$

Indeed, Theorem 3(a) bounds survival by a constant times the wedge harmonic function at a fixed inward shift, divided by $k^{p/2}$. The harmonic function is homogeneous of degree p and is bounded above by a constant times $\|z\|^p$, giving (5.1). Theorem 3(b) gives a positive integral over the ball in (5.2); Theorem 2 and accessibility give positivity of the starting harmonic function.

All hypotheses can be checked for the present skeletons. The moment and covariance conditions follow from (3.10). A planar wedge is Lipschitz and star-like. In polar coordinates its positive Brownian harmonic function is

$$u(r, \varphi) = r^p \sin(p\varphi), \quad 0 < \varphi < \theta, \quad (5.3)$$

which is comparable to r^{p-1} times distance to the boundary. The accessibility constructions in Section 4.1 supply the positive starting points used in the lower bound.

The lattice boundary convention causes no difficulty, but must be fixed before whitening. On the even lattice, translating (a, b) to $(a + 1, b + 1)$ makes $a, b \geq 0$ equivalent to positivity of both translated coordinates. Translating to $(a - 1, b - 1)$ does the same for $a, b \geq 2$. For Q_S , translation by $(1, 0)$ makes $a \geq 0, b \geq 1$ equivalent to an open quadrant condition on either parity class. Each is a fixed shift. After diffusive rescaling, the induced displacement of endpoint balls tends to zero, and nested fixed balls suffice in every application of (5.2).

6 Transfer to exact model time

This section applies to both models and their reversals. For P , time is physical time; for S , it is aggregate time. Put

$$S_n(a) = \sum_{b \in Q} q_n(a, b) = \mathbb{P}_a(X_t \in Q \text{ for } 0 \leq t \leq n). \quad (6.1)$$

Proposition 6.1 (Survival upper bound). *For the starting parity being used for regeneration,*

$$S_n(a) \leq C(1 + \|a\|)^p n^{-p/2} + Ce^{-cn}. \quad (6.2)$$

The constants can be chosen for all starting points in the relevant domain and either of the two parity classes.

Proof. Take $k = \lfloor n/4 \rfloor$. Since the iid durations have mean two and an exponential moment, a Chernoff bound gives $\mathbb{P}(T_k > n) \leq Ce^{-cn}$. If the model-time path survives through n and $T_k \leq n$, then all of the first k skeleton endpoints lie in Q . Drop all within-cycle restrictions and apply (5.1), with the appropriate fixed translation. This is an event inclusion, not an identification of skeleton survival with full path survival. There are only two parities, so their constants can be combined. $\square$

Proposition 6.2 (Interior mass at every exact time). *For each ball B with closure in the positive quadrant, a constant $c_B > 0$ exists such that*

$$\mathbb{P}_{(2,2)}(X_t \in Q \ (0 \leq t \leq n), \ X_n \in \sqrt{n}B) \geq c_B n^{-p/2} \quad (6.3)$$

for all sufficiently large integers n . This is true for the forward and reverse chains in both models.

Proof. Choose a smaller ball B' with $\overline{B'} \subset B$ and set

$$k = \lceil n/2 + n^{3/4} \rceil. \quad (6.4)$$

Constrain the even skeleton endpoints to have both coordinates at least two. Equation (5.2), after the fixed translation, gives

$$\mathbb{P}(X_{T_j} \in [2, \infty)^2 \ (0 \leq j \leq k), \ X_{T_k} \in \sqrt{n}B') \geq cn^{-p/2}. \quad (6.5)$$

To justify the varying normalization, note that $k/n \rightarrow 1/2$. A sufficiently small fixed ball whose closure lies in $\sqrt{2}B'$ is contained in the rescaled target for all large n , even after the fixed lattice shift. Thus only a fixed-ball application of (5.2) is needed.

The mean of T_k is $2k = n + 2n^{3/4} + O(1)$. Exponential moments of the durations give

$$\mathbb{P}\{T_k \notin [n, n + 4n^{3/4}]\} \leq Ce^{-cn^{1/2}}. \quad (6.6)$$

For completeness, the usual moment-generating-function bound for a centered duration is $\mathbb{E}e^{\lambda(\ell-2)} \leq e^{C\lambda^2}$ for sufficiently small $|\lambda|$. Choosing λ proportional to $n^{-1/4}$ in a deviation of order $n^{3/4}$ over $k = O(n)$ terms proves (6.6).

For the corrected model-time martingale, the conditional exponential moments are uniformly bounded over the two states. Since each increment has conditional mean zero, Taylor expansion gives

$$\mathbb{E}(e^{\lambda \Delta M_t^{(r)}} \mid \mathcal{F}_t) \leq e^{C\lambda^2} \quad (|\lambda| \leq \lambda_0, \ r = 1, 2). \quad (6.7)$$

The exponential supermartingale maximal inequality, applied to both signs and both coordinates, implies for fixed $\delta > 0$

$$\mathbb{P}\left(\max_{0 \leq r \leq \lceil 4n^{3/4} \rceil + 2} \|X_{n+r} - X_n\| > \delta\sqrt{n}\right) \leq Ce^{-c_\delta n^{1/4}}. \quad (6.8)$$

One can take λ proportional to $\sqrt{n}/n^{3/4} = n^{-1/4}$; the bounded corrector changes thresholds by only a constant. This estimate is unconditional and uniform in the state at time n . The reverse corrector verified above gives the identical argument in reverse time.

On the event in (6.5), Lemma 4.1 gives survival through T_k . Remove the bad events in (6.6) and (6.8). If δ is smaller than the separation of $\overline{B'}$ from B^c , then $X_n \in \sqrt{n}B$ and the path survives through the exact time n . Both subtracted errors are exponentially smaller than $n^{-p/2}$. This proves (6.3) without any conditioning of clock or displacement estimates on survival. $\square$

The entrance paths in Section 4.1 extend (6.3) to the fixed endpoints needed below. A smaller target ball absorbs their fixed time lengths. Explicitly, for P the forward starting point is $(0, 0)$ and the reverse starting point is $(4, 0)$; for S they are $(0, 2)$ and $(1, 1)$, respectively. The resulting endpoint masses are each bounded below by a positive constant times $n^{-p/2}$.

7 Free local bounds and lattice aperiodicity

Represent every $z \in L$ uniquely as

$$z = (2a + j, 2b + j), \quad (a, b) \in \mathbb{Z}^2, \quad j \in \{0, 1\}. \quad (7.1)$$

This is a two-state translation-cell representation. Let $F(t)$ be the Fourier matrix of its cell increments, with $t \in [-\pi, \pi]^2$. If $D(t) = \text{diag}(1, e^{i(t_1+t_2)/2})$, then

$$F(t) = D(t)K(e^{it_1/2}, e^{it_2/2})D(t)^{-1}. \quad (7.2)$$

Indeed, a transition from internal state i to j with physical displacement z has cell displacement $(z - (j - i)(1, 1))/2$. The direct cell formula also makes F 2π -periodic in each coordinate.

Lemma 7.1 (No secondary Fourier points). *The only point of the cell torus where $F(t)$ has spectral radius one is $t = 0$. At zero the sole unit-modulus eigenvalue is the simple eigenvalue one.*

Proof. Suppose $F(t)v = \lambda v$ with $|\lambda| = 1$, and normalize the largest component modulus of v to one. Equality in the triangle inequality, together with irreducibility of the internal stochastic matrix, forces both component moduli to equal one and all supported transition phases to align. In particular, all $e \rightarrow e$ cell-increment phases are equal. For P their support includes every $(-i, j)$ with $i \geq 0, j \geq 1$. Comparing $(0, 1)$ and $(0, 2)$ forces $t_2 = 0$ modulo 2π ; comparing $(0, 1)$ and $(-1, 1)$ forces $t_1 = 0$. For S , the support includes every $(-i, j)$ with $i \geq 1, j \geq 0$. Comparing $(-1, 0), (-2, 0), (-1, 1)$ gives the same conclusion. At zero, the matrix in (3.1) is primitive; its other eigenvalue is $2a - 1$, strictly inside the unit circle. $\square$

Proposition 7.2 (Free local estimates). *For either chain, with no killing,*

$$\sup_{a, b \in L} \mathbb{P}_a(X_n = b) \leq Cn^{-1}. \quad (7.3)$$

For every fixed $A < \infty$, uniformly for $a, b \in L$ with $\|b - a\| \leq A\sqrt{n}$,

$$\mathbb{P}_a(X_n = b) = \frac{2}{2\pi n \sqrt{\det \Sigma}} \exp\left(-\frac{(b - a)^T \Sigma^{-1} (b - a)}{2n}\right) + o(n^{-1}). \quad (7.4)$$

The statements hold for all four initial and final parity pairs, and for the reverse chains.

Proof. Near zero the principal eigenvalue and spectral projection of $F(t)$ are analytic. Centering and (3.7) give

$$\lambda(t) = 1 - \frac{1}{2}t^T(\Sigma/4)t + O(\|t\|^3), \quad \Pi(t) \longrightarrow \mathbf{1}(1/2, 1/2). \quad (7.5)$$

To identify the Hessian without presuming an iid walk at model time, write $X_n = M_n - h_{J_n}$. Martingale increment orthogonality and convergence of the finite internal chain give $n^{-1} \text{Cov}(M_n) \rightarrow \Sigma$; the bounded corrector changes the covariance divided by n by $o(1)$. Analytic differentiation of the finite matrix moment-generating function identifies the same limiting covariance with the Hessian of the log principal eigenvalue. Passing to cells divides it by four. The mean is zero at linear scale for the same reason.

In a small neighborhood of zero, $|\lambda(t)| \leq e^{-c\|t\|^2}$ and the other spectral part decays exponentially in n . On the compact complement, Lemma 7.1 gives a uniform strict spectral-radius bound, and finite-dimensional matrix-power estimates give exponential decay. Fourier inversion therefore proves (7.3). Rescaling t by $n^{-1/2}$ in the neighborhood of zero gives the Gaussian local limit, uniformly on bounded diffusive displacement ranges. The cell covariance contributes a factor four when expressed in physical coordinates; the terminal parity mass is $1/2$. Their product is the factor two in (7.4), also the covolume of L in $\mathbb{Z}^2$. Bounded parity-coordinate offsets change the exponent by $o(1)$. Reversal preserves the same argument. $\square$

This Fourier calculation supplies the exact-time aperiodicity that is not supplied merely by an invariance principle. There is no leftover restriction to even times.

8 A killed interior estimate and gluing

Lemma 8.1 (Interior killed local lower bound). *Fix $\eta > 0$. For a sufficiently large fixed R , set*

$$B = B((R, R), \eta), \quad B_n = \sqrt{n}B \cap L. \quad (8.1)$$

There is a constant $c > 0$ such that, for all sufficiently large n and every integer $m \in [n/4, n/2]$,

$$q_m(a, b) \geq c/n \quad (a, b \in B_n). \quad (8.2)$$

Proof. Equation (7.4) gives $\mathbb{P}_a(X_m = b) \geq c_0/n$ uniformly in the specified range. Here c_0 is independent of R , since the Gaussian depends on $b - a$, whose norm is at most $2\eta\sqrt{n}$.

Let τ be the first exit from Q . For paths that exit in the first half of the interval, the strong Markov property at τ and (7.3) give

$$\mathbb{P}_a(X_m = b, \tau \leq \lfloor m/2 \rfloor) \leq \frac{C}{n} \mathbb{P}_a(\tau \leq \lfloor m/2 \rfloor). \quad (8.3)$$

Every starting point in B_n is at coordinate distance at least $(R - \eta)\sqrt{n} - O(1)$ from the complement of Q . The bounded corrector and Doob's quadratic martingale maximal inequality yield

$$\mathbb{P}_a(\tau \leq \lfloor m/2 \rfloor) \leq \frac{C}{(R - \eta)^2}. \quad (8.4)$$

The expected martingale quadratic variation is $O(m)$ by the finite matrices (3.5)–(3.6). For a path that first exits in the second half, reverse the full unrestricted path from its fixed endpoint b . Some reversed position outside Q is then visited in the first half of reversed time. Equation (3.12) and the same argument bound this mass by $C/[n(R - \eta)^2]$. Taking a union bound covers all paths with any exit. Choosing R large makes this smaller than $c_0/(2n)$; subtracting from the free lower bound proves the claim. The fixed shift of Q_S changes only the harmless $O(1)$ boundary distance. $\square$

No cone-killed local limit theorem or conditioned bridge theorem is used in this step.

8.1 Upper bounds

Split $n = m + \ell + m$, where $m = \lfloor n/3 \rfloor$ and $\ell = n - 2m$. The Markov property, the unrestricted middle-kernel upper bound, and reversal give

$$\begin{aligned} q_n(a, b) &= \sum_{u, v \in Q} q_m(a, u) q_\ell(u, v) q_m(v, b) \\ &\leq \frac{C}{n} S_m(a) S_m^*(b). \end{aligned} \quad (8.5)$$

For P , Proposition 6.1 gives

$$q_n^P((0, 0), (2i, 0)) \leq C(1 + i)^{p_P} n^{-p_P - 1}. \quad (8.6)$$

Exponential clock errors are absorbed in this bound uniformly in i . Summation in (2.6) is legitimate because $\sum_{i \geq 1} 3^{-i} (1 + i)^{p_P} < \infty$. This proves the upper half of the P estimate.

For S , apply (8.5) to the fixed endpoints $a = (0, 2)$ and $b = (1, 1)$. Regenerate at even parity from a and at odd parity for the reverse chain from b . Equations (3.10) and (6.2) apply in both cases, so

$$q_N^S((0, 2), (1, 1)) \leq C N^{-p_S - 1}. \quad (8.7)$$

8.2 Lower bounds

Use the same three-piece split and restrict both joining points u, v to B_n from Lemma 8.1. The entrance paths and Proposition 6.2 give forward and reverse masses at least $cn^{-p/2}$ in B_n . To handle the floor in m , choose a fixed ball C with closure inside $\sqrt{3}B$. Since $\sqrt{n/m} \rightarrow \sqrt{3}$, one has $\sqrt{m}C \subset \sqrt{n}B$ for every sufficiently large n . A fixed-ball use of Proposition 6.2 at time m supplies the required mass. The fixed entrance times are handled by one further nested ball, if needed.

The middle length ℓ lies in $[n/4, n/2]$ for all large n . Thus

$$q_n(a, b) \geq \sum_{u, v \in B_n} q_m(a, u) \frac{c}{n} q_m^*(b, v) \geq cn^{-p/2} n^{-1} n^{-p/2} = cn^{-p-1} \quad (8.8)$$

for the endpoint pairs $(a, b) = ((0, 0), (4, 0))$ in P and $((0, 2), (1, 1))$ in S .

Keeping only $i = 2$ in (2.6) proves $p_n \geq c\mu_P^n n^{-p_P - 1}$. For S , (2.13) gives $s'_n = \Theta(\mu_S^n n^{-p_S - 1})$; then (2.8), with nonnegative summands and fixed shifts of n , gives the same estimate for s_n . This completes the proof of Theorem 1.1.

9 Consequences

9.1 Irrational exponents and non-D-finiteness

Lemma 9.1. *Both α_P and α_S are irrational.*

Proof. Let $r \in \{9/16, 22/27\}$ and suppose $1 + \pi / \arccos r$ is rational. Then $\arccos r$ is a rational multiple of π , so $\zeta = e^{i \arccos r}$ is a root of unity. Therefore $2r = \zeta + \zeta^{-1}$ is an algebraic integer. As a rational algebraic integer it must be an integer, contradicting $2r = 9/8$ or $44/27$. $\square$

Lemma 9.2 (A two-sided coefficient criterion). *Suppose a_n are nonnegative integers, $\mu > 1$, α is irrational, and $a_n = \Theta(\mu^n n^{-\alpha})$. Then $A(z) = \sum_{n \geq 0} a_n z^n$ is not D-finite.*

Proof. Put $\rho = \mu^{-1}$ and choose a nonnegative integer $k > \alpha - 1$. The coefficient bounds and nonnegativity imply, as $t \downarrow 0$,

$$A^{(k)}(\rho(1-t)) = \Theta\left(\sum_{n \geq k} n^{k-\alpha}(1-t)^n\right) = \Theta(t^{\alpha-k-1}). \quad (9.1)$$

Factors $\rho^{-k}(1-t)^{-k}$ are bounded and positive, and the falling factorial $n(n-1)\cdots(n-k+1)$ is comparable to n^k for large n . With $\beta = k - \alpha > -1$, the last estimate follows from

$$\sum_{n \geq 1} n^\beta e^{-sn} \sim \Gamma(\beta+1)s^{-\beta-1}, \quad s \downarrow 0, \quad (9.2)$$

by a Riemann-sum comparison and $s = -\log(1-t) \sim t$. Initial terms do not affect a divergent derivative.

If A were D-finite, its integer coefficients and exponential upper bound would make it a G -function: its coefficient conjugates have exponential growth and its coefficient denominators are all one. The André–Chudnovsky–Katz regular-singularity and rational-exponent theorem implies that its minimal differential equation has only regular singularities with rational local exponents. The finite rational-power and logarithmic form used below is stated explicitly in [8, Theorem 6, Section 4.1]; see also [9, Section 2.2, proof of Theorem 2].

Consequently, a local solution near $z = \rho$, and every fixed derivative of it, has a finite Frobenius expansion in rational powers and nonnegative integer powers of a logarithm, with analytic coefficient functions. Along the positive real approach, a nonzero divergent such function has a rational logarithmic growth exponent: combine terms with equal powers, choose the smallest surviving rational power, and then the highest nonzero logarithmic power at that order. In contrast, (9.1) gives

$$\lim_{t \downarrow 0} \frac{\log A^{(k)}(\rho(1-t))}{\log(1/t)} = k + 1 - \alpha, \quad (9.3)$$

which is irrational. This contradiction proves the result. $\square$

Remark 9.3. The coefficient criterion stated in [9, Theorem 2] assumes an asymptotic equivalent. It is not directly applicable to a Θ estimate. The positive-derivative argument in (9.1) is the additional step used here. For a rational-coefficient series, D-finiteness over $\mathbb{C}$ is equivalent to D-finiteness over $\mathbb{Q}$: a relation of fixed order and polynomial degree imposes rational linear equations on finitely many operator coefficients, and a rational row-space basis gives a nonzero rational solution whenever a nonzero complex solution exists.

[**write, 2026-10-05**] The two-sided bounds are more than the argument needs: with a derivative order $k > \alpha - 1$ and an ε of room in the exponent, the same proof works for $a_n = \mu^n n^{-\alpha+o(1)}$. This form is run in Part III (Section 27) and stated in general in Remark 27.1, and also in Remark 10.1 (`osr:rem:criterion`) of the report `a348351-one-sided-rectangulations` in this collection (bundle Report 111, batch 103). There the irrationality certificate differs from Lemma 9.1: $2 \cos \theta = (-29 + 7\sqrt{17})/2$ is an algebraic integer, and the certificate is its conjugate, which lies below -2 .

Corollary 9.4. *The ordinary generating functions of A377922, A377920, and A377921 are not D-finite.*

Proof. The first two cases follow from Theorem 1.1 and Lemmas 9.1–9.2. For A377921, let $\tilde{s}_n$ be the rigid-orthogonal-surface sequence, and put $S(z) = \sum_{n \geq 2} s_n z^n$, $T(z) = \sum_{n \geq 2} \tilde{s}_n z^n$. Remark 20 of [1] gives

$$S(z) = (1+z)^3 T(z) + 3z^2 + z^3. \quad (9.4)$$

D-finite series are closed under multiplication by rational functions and addition of polynomials. Equation (9.4) therefore makes S D-finite if and only if T is D-finite. $\square$

No coefficientwise Θ estimate for A377921 follows from this argument; the statement about that sequence concerns its generating function only.

[write, 2026-10-02] A coefficientwise statement for A377921 is now proved in Part II: equation (11.4) gives $\tilde{s}_n \sim (16/19)^3 \kappa_S \mu_S^n n^{-\alpha_S}$, by a dominated signed convolution that uses the full equivalent for s_n . As Part II says, the Θ bounds of Theorem 1.1 alone do not give it.

[write, 2026-10-05] A coefficientwise Θ estimate for A377921 does follow from Theorem 1.1 together with the sleeve injection of Part III, which gives $s_{n-6} \leq \tilde{s}_n \leq s_n$ for $n \geq 8$ (Proposition 29.1, Remark 29.3). Part III also proves non-D-finiteness from the logarithmic law alone (Section 27, Remark 27.1), a weaker hypothesis than that of Lemma 9.2.

9.2 First-threshold inverse asymptotics

Lemma 9.5. *Suppose $a_n = \Theta(\mu^n n^{-\alpha})$ with $\mu > 1$ and $\alpha \in \mathbb{R}$, and define $N(Y) = \min\{n \geq 0 : a_n \geq Y\}$. Then*

$$N(Y) = \frac{\log Y + \alpha \log \log Y}{\log \mu} + O(1) \quad (Y \rightarrow \infty). \quad (9.5)$$

The sequence a_n need not be monotone.

Proof. Put $L = \log Y$ and $v(L) = (L + \alpha \log L)/\log \mu$. Uniformly for h in a fixed bounded interval,

$$\log(\mu^{v(L)+h} (v(L) + h)^{-\alpha}) = L + h \log \mu + \alpha \log \log \mu + o(1). \quad (9.6)$$

Choose a constant H sufficiently large that the upper coefficient comparison is less than Y at $v(L) - H$, while the lower comparison is at least Y at $v(L) + H$, allowing another fixed unit for integer rounding. The comparison function $\mu^n n^{-\alpha}$ is eventually increasing. Hence all sufficiently large integers below the lower bracket have $a_n < Y$, and the finitely many earlier values also do so for large Y . The upper bracket supplies at least one index above threshold. This proves (9.5). $\square$

In particular,

$$N_P(Y) = \frac{\log Y + \alpha_P \log \log Y}{\log(9/2)} + O(1), \quad (9.7)$$

$$N_S(Y) = \frac{\log Y + \alpha_S \log \log Y}{\log(16/3)} + O(1). \quad (9.8)$$

These estimates do not specify an exact rounding rule or a next additive constant.

[write, 2026-10-02] Part II, Theorem 20.1, sharpens (9.7)–(9.8) under the full equivalents: Lambert- W_{-1} centering with a next additive constant $-(\alpha \log \log \mu + \log \gamma)/\log \mu$ and a vanishing-width two-ceiling bracket. It still gives no exact rounding rule. Both inversions are instances of the repository’s transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`): the model equation $\lambda r - \alpha \log r = \log(Y/\gamma)$ is `p0:thm:lambert-core` with $a = \log \mu$, $b = -\alpha$ (branch W_{-1}), its expansion is the first step of `p0:thm:lambert-centered`, and the integer bracket is the separation situation of `p0:thm:staircase`. No novelty is claimed for the inversion mechanics.

10 Verification and unresolved questions

The reproducibility package contains editable L^AT_EX source, the compiled PDF, exact symbolic and enumeration programs, their recorded outputs, source identifiers, and mathematical audit receipts. Appendix A specifies the finite checks. These checks corroborate normalizations and identities; they are not a formal verification of the probabilistic proof or an external peer review.

[write, 2026-10-02] In this repository the delivered PDF and the checksum ledger of this package are not shipped; the report’s `article.pdf` is a build of this merged text. The programs, outputs, source excerpts and receipts are shipped byte-identical under the prefix `45-cone-` (programs in `code/`, outputs and data in `data/`, the two audit notes at the report root). The report README lists them and gives a rerun recipe.

Theorem 1.1 does not prove either of the limits

$$\lim_{n \rightarrow \infty} p_n \mu_P^{-n} n^{\alpha_P}, \quad \lim_{n \rightarrow \infty} s_n \mu_S^{-n} n^{\alpha_S}. \quad (10.1)$$

It proves that both normalized sequences are eventually bounded above and bounded away from zero. Convergence to positive constants, the full asymptotic equivalents of [1, Conjecture 25], remains to be established. Computing those constants and correction terms would require sharper information than the inequalities used in the clock transfer and gluing arguments. Other open directions include a direct coefficientwise estimate for A377921 and identifying other parity-dependent tandem models for which a fixed cycle buffer exists.

[write, 2026-10-02] Status after the merge. Convergence to positive constants and the full equivalents of [1, Conjecture 25] are proved in Part II (Theorem 11.1), and so is a direct coefficientwise equivalent for A377921 (equation (11.4)). Still open: the numerical values of κ_P and κ_S , a rate of convergence, correction terms, an exact rounding rule for the threshold inverses, and other parity-dependent tandem models with a fixed cycle buffer.

[write, 2026-10-05] For the last item, Part III’s Theorem 24.1 gives the logarithmic-scale exponent for every walk with a finite internal state that satisfies (H1)–(H5), with or without a fixed cycle buffer; bundle Report 111 (`a348351-one-sided-rectangulations`, same collection directory) is one such chain (Section 22.5). Models with a buffer still matter for amplitudes and Θ bounds, which the general theorem does not give.

This research note presents a mathematical proof with reproducible exact checks. It has not been externally peer reviewed or machine-formalized, and makes no publication-priority claim. The source credits distinguish the established bijections and iid cone input from the argument presented here.

Part II

Limiting amplitudes (source 46)

Limiting amplitudes for corner polyhedra and Schnyder labelings

Research addendum, 2 October 2026

Abstract of source 46, as delivered. We strengthen the two-sided estimates for the parity-dependent quadrant walks counting corner polyhedra and 3-connected Schnyder labelings to full asymptotic equivalents with finite positive amplitudes. The key step is a harmonic comparison for a cycle kernel trapped between an ordinary iid cone kernel and its fixed inward translate. It yields exact survival constants and a common endpoint limit law for valid regeneration cycles. A two-sided random-clock argument transfers these limits to every exact counted time. A killed local limit, proved from the free local theorem and the exit decomposition, then gives the fixed-endpoint amplitudes by a fixed-thirds semigroup limit. Geometrically weighted endpoint summation and a signed rational convolution give the enumerative equivalents, including the rigid-surface sequence. The amplitude constants have positive harmonic and heat-kernel expressions; no numerical digits are certified. We also give Lambert- W inverse centering and a vanishing-width ceiling bracket, without asserting an unconditional rounding formula.

11 Results and the accompanying foundation

This addendum is intended to be read with the included research note *Polynomial exponents for corner polyhedra and Schnyder labelings*, cited as the *Foundation* (in this report: Part I). The Foundation and its exact-check package are included unchanged. They establish the walk models, covariance calculations, fixed-buffer geometry, free local theorem and two-sided estimates used below. This addendum supplies the limiting-amplitude argument, rather than assuming a cone local theorem for the original parity-dependent chain. Statements in the Foundation that amplitude convergence was unresolved describe the scope of that earlier note; the present addendum establishes those limits.

[write, 2026-10-02] In this report the Foundation is Part I (Sections 1–10 and Appendix A), printed with source 45’s numbering, so that “Foundation Lemma 2.1” below is Lemma 2.1 here; the Foundation references are hyperlinks. The unchanged 30-file copy of the Foundation package that source 46 embedded is not shipped: the same files are shipped once, from source 45, with the prefix 45-cone- (the copy’s PDF and checksum ledger only in the arrival commit). The source’s Section n is Section $n + 10$ here.

Let p_n count polyhedral orientations with n inner vertices, equivalently corner-polyhedron types with $n + 3$ flats. Let s_n count 3-connected Schnyder labelings with n inner faces, and let $\tilde{s}_n$ be the rigid-orthogonal-surface sequence of Fusy–Narmanli–Schaeffer [1]. These are A377922, A377920 and A377921, respectively. Set

$$\mu_P = \frac{9}{2}, \quad \mu_S = \frac{16}{3}, \quad \alpha_P = 1 + \frac{\pi}{\arccos(9/16)}, \quad \alpha_S = 1 + \frac{\pi}{\arccos(22/27)}. \quad (11.1)$$

Theorem 11.1 (Full enumerative equivalents). *There are finite strictly positive constants κ_P and*

κ_S such that

$$p_n \sim \kappa_P \mu_P^n n^{-\alpha_P}, \quad (11.2)$$

$$s_n \sim \kappa_S \mu_S^n n^{-\alpha_S}, \quad (11.3)$$

$$\tilde{s}_n \sim \left(\frac{16}{19}\right)^3 \kappa_S \mu_S^n n^{-\alpha_S}. \quad (11.4)$$

The limits hold on the full sequence of integers. Positive expressions for the amplitudes are given in Section 19.

The first two conclusions are the positive-amplitude equivalents stated in [1, Conjecture 25]. The third follows here by a dominated signed-convolution argument, which uses the full equivalent and is not a consequence of the earlier Θ bounds alone.

Sequence	Rate	Power	Amplitude
A377922	9/2	α_P	$\kappa_P > 0$
A377920	16/3	α_S	$\kappa_S > 0$
A377921	16/3	α_S	$(16/19)^3 \kappa_S$

No numerical evaluation of these constants, convergence rate, or further coefficient correction is claimed. This is a mathematical research addendum with exact reproducibility checks, not an externally peer-reviewed publication or a machine-formalized proof. No publication-priority claim is made.

12 Model notation and established inputs

[**write, 2026-10-02**] This section restates inputs proved or quoted in Part I, in source 46's notation; it is kept because the rest of Part II cites its displays. D_P and H_S here are Part I's D and H ((2.4), (2.11)); the letter D alone is reserved in this Part for the valid-cycle kernel of Section 13. Table 1 lists every clash between the two Parts.

Let

$$L = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod{2}\}, \quad C = (0, \infty)^2. \quad (12.1)$$

For P , one unit of counted time is one physical tandem step and the killed domain is $Q_P = L \cap [0, \infty)^2$. For S , one unit is an aggregate consisting of a southeast step $(1, -1)$ followed by zero or more weighted face steps, and the killed domain is $Q_S = \{(a, b) \in L : a \geq 0, b \geq 1\}$. In the latter model, restricting aggregate endpoints to Q_S controls all physical intermediate steps, as proved in Foundation Lemma 2.1.

Write $q_n(a, b)$ for the counted-time transition probability killed outside the model domain. The model is specified by a 2×2 probability-generating matrix; its two internal states are the parity of the ordinate. In the notation

$$D_P = \left(1 - \frac{1}{3x^2}\right) \left(1 - \frac{y^2}{3}\right), \quad H_S = 1 - x^{-2} - y^2 - 2x^{-2}y^2, \quad (12.2)$$

the matrices are

$$K_P(x, y) = \begin{pmatrix} 2y^2/(27D_P) & 2x/(3y) + 2y/(27xD_P) \\ 2x/(3y) + 2y/(27xD_P) & 2/(27x^2D_P) \end{pmatrix}, \quad (12.3)$$

$$K_S(x, y) = \frac{3}{16}M_S(2x, y/2), \quad M_S(x, y) = \begin{pmatrix} x^{-2}/H_S & x/y + x^{-1}y/H_S \\ x/y + x^{-1}y/H_S & y^2/H_S \end{pmatrix}. \quad (12.4)$$

The covariance matrices per unit counted time are

$$\Sigma_P = \begin{pmatrix} 16/5 & -9/5 \\ -9/5 & 16/5 \end{pmatrix}, \quad \Sigma_S = \begin{pmatrix} 36/7 & -88/21 \\ -88/21 & 36/7 \end{pmatrix}. \quad (12.5)$$

Counting measure on L is invariant. The reversed chain has $K_{ij}^*(z) = K_{ji}(-z)$ and satisfies

$$q_n(a, b) = q_n^*(b, a). \quad (12.6)$$

The symmetric internal matrices have stationary distribution $(1/2, 1/2)$.

For either initial parity, let T_k be successive first-return times to that parity. The cycle pairs (Y_k, ℓ_k) of displacement and duration are iid, with

$$\mathbb{E}Y = 0, \quad \text{Cov}(Y) = 2\Sigma, \quad \mathbb{E}\ell = 2, \quad \mathbb{E}e^{\varepsilon(\|Y\|+\ell)} < \infty \quad (12.7)$$

for some $\varepsilon > 0$. The reversed cycles have displacement law $-Y$ and the same duration law. Dependence between displacement and duration is permitted and retained.

The following previously established estimates will be used with their stated scopes:

1. *Cycle buffer.* Every intermediate counted position in a first-return cycle with endpoints a, b has coordinate r at least $\min(a_r, b_r) - 1$. This holds forward and backward and at either parity (Foundation Lemma 4.1).
2. *Bounded corrector.* $M_n = X_n + h_{J_n}$ is a martingale; its increments have uniformly bounded conditional exponential moments. The reversed chain has corrector $-h$ and the same averaged covariance (Foundation Section 3).
3. *Survival upper bound.* For either chain,

$$\mathbb{P}_a(\tau > n) \leq C(1 + \|a\|)^p n^{-p/2} + Ce^{-cn}, \quad (12.8)$$

where $p = \alpha - 1 > 1$ (Foundation Proposition 6.1).

4. *Free local theorem.* On the unrestricted chain,

$$\sup_{a, b \in L} \mathbb{P}_a(X_n = b) \leq C/n, \quad (12.9)$$

and, uniformly on compact diffusive displacement sets,

$$\mathbb{P}_a(X_n = b) = \frac{2}{2\pi n \sqrt{\det \Sigma}} \exp\left(-\frac{(b-a)^T \Sigma^{-1}(b-a)}{2n}\right) + o(n^{-1}). \quad (12.10)$$

All parity pairs and all sufficiently large integer times are included. The factor two is the covolume of L (Foundation Proposition 7.2).

5. *Uniform endpoint estimate for P .*

$$q_n^P((0, 0), (2i, 0)) \leq C(1 + i)^{p_P} n^{-p_P-1}, \quad i \geq 1 \quad (12.11)$$

(Foundation equation (8.6)).

The normalization identities are

$$p_n = \mu_P^n \sum_{i \geq 1} 3^{-i} q_n^P((0, 0), (2i, 0)), \quad (12.12)$$

$$s'_n = \frac{\mu_S^{n+1}}{4} q_{n+1}^S((0, 2), (1, 1)), \quad s_n = s'_n + 2s'_{n-1} + s'_{n-2} \quad (n \geq 4), \quad (12.13)$$

where s'_n restricts the two outer G vertices to be non-isolated. The full rigid-series relation is

$$S(z) = (1+z)^3 \tilde{S}(z) + 3z^2 + z^3, \quad S(z) = \sum_{n \geq 2} s_n z^n, \quad \tilde{S}(z) = \sum_{n \geq 2} \tilde{s}_n z^n. \quad (12.14)$$

These are the counting bijections and relations of [1, Propositions 13–14, Remarks 9 and 20], in the normalized form verified in the Foundation.

13 Ordinary and valid-cycle cone kernels

Fix a model, an initial parity and a time direction. Let

$$T = (2\Sigma)^{-1/2}, \quad K = TC. \quad (13.1)$$

Whitening the cycle covariance makes its increment covariance the identity. The fixed translation of the original domain is $c_P = (1, 1)$ or $c_S = (1, 0)$, before applying T . A starting point a thus becomes $x = T(a + c_M)$, for model M . The open wedge K has angle

$$\theta = \arccos\left(-\frac{\Sigma_{12}}{\sqrt{\Sigma_{11}\Sigma_{22}}}\right), \quad p = \pi/\theta > 1. \quad (13.2)$$

In polar coordinates adapted to its boundary rays, normalize the positive Brownian harmonic function by

$$u(r, \varphi) = r^p \sin(p\varphi), \quad 0 < \varphi < \theta. \quad (13.3)$$

All kernels below live on the corresponding translated and whitened lattice coset. Fixed translations need not preserve that coset to define the translated starting state of the same iid walk.

Let P_0 be the iid cycle-displacement kernel killed outside K . Choose

$$b_P = T(2, 2), \quad b_S = T(2, 1), \quad K_b = b + K. \quad (13.4)$$

On either parity lattice, the inner condition $T(a + c_M) \in K_b$ is exactly $a_1, a_2 \geq 2$. Let P_b be the iid displacement kernel killed outside K_b , extended by zero for starting points outside K_b . Let D be the true substochastic kernel of one complete cycle, rejecting a cycle if any of its counted steps leaves the model domain. By the cycle buffer,

$$0 \leq P_b \leq D \leq P_0 \quad (13.5)$$

pointwise as kernels. In fact, if both endpoints lie in K_b , every marked cycle between them is safe, so D agrees there with the unrestricted cycle kernel. The inequality concerns entire kernels, not only their row sums.

13.1 The iid harmonic input

The hypotheses of Denisov–Wachtel [2, Theorems 2–3] hold for these iid increments: centering, identity covariance, exponential moments, the wedge geometry and $p \geq 1$ were checked in the Foundation. Write Z_m for the whitened iid walk, τ_0 for its exit from K , and

$$V(x) = \lim_{m \rightarrow \infty} \mathbb{E}_x[u(Z_m); \tau_0 > m]. \quad (13.6)$$

The theorem provides $P_0 V = V$, uniform $V(z)/u(z) \rightarrow 1$ as $\text{dist}(z, \partial K) \rightarrow \infty$, a polynomial survival upper bound, and

$$m^{p/2} P_0^m \mathbf{1}(x) \rightarrow \beta V(x), \quad (13.7)$$

$$m^{p/2} \mathbb{E}_x[f(Z_m/\sqrt{m}); \tau_0 > m] \rightarrow \beta V(x) \nu(f), \quad (13.8)$$

for bounded continuous nonnegative f . Here $\beta > 0$ uses the normalization (13.3), and ν is the normalized Brownian cone-meander endpoint law in K . The corresponding translated kernel has harmonic function

$$V_b(x) = V(x - b) \mathbf{1}_{\{x \in K_b\}}, \quad P_b V_b = V_b, \quad (13.9)$$

and the same constants β, ν , since the shift is fixed. Coupling the same iid increments started at x and $x - b$ makes survival in the translated inner wedge imply survival in the outer wedge. Taking the limits in (13.7) gives

$$0 \leq V_b(x) \leq V(x). \quad (13.10)$$

14 The boundary-shift comparison

Lemma 14.1. *For every fixed outer starting point x ,*

$$P_0^m (V - V_b)(x) \rightarrow 0. \quad (14.1)$$

Proof. Since $P_0^m V(x) = V(x)$, it suffices to prove $P_0^m V_b(x) \rightarrow V(x)$. Put $d(z) = \text{dist}(z, \partial K)$. We first show that for every fixed R ,

$$\mathbb{E}_x[u(Z_m); \tau_0 > m, d(Z_m) \leq R] \rightarrow 0. \quad (14.2)$$

On $\|Z_m\| \leq \sqrt{m} \log m$, the wedge estimate $u(z) \leq C\|z\|^{p-1}d(z)$ and the survival upper bound give

$$\mathbb{E}_x[u(Z_m); \tau_0 > m, d(Z_m) \leq R, \|Z_m\| \leq \sqrt{m} \log m] \leq C_x R m^{-1/2} (\log m)^{p-1} = o(1). \quad (14.3)$$

For the complementary region no killing estimate is needed. Centered iid exponential moments imply, for t sufficiently larger than the fixed starting norm,

$$\mathbb{P}_x(\|Z_m\| > t) \leq C_x \exp[-c \min(t^2/m, t)]. \quad (14.4)$$

For $L_m = \sqrt{m} \log m$, integration of this tail bound yields

$$\begin{aligned} \mathbb{E}_x[\|Z_m\|^p; \|Z_m\| > L_m] &\leq L_m^p \mathbb{P}_x(\|Z_m\| > L_m) + \int_{L_m}^{\infty} p t^{p-1} \mathbb{P}_x(\|Z_m\| > t) dt \\ &= O\left(m^{p/2} e^{-c'(\log m)^2}\right) + O(e^{-c'm}) = o(1), \end{aligned} \quad (14.5)$$

after absorbing powers of $\log m$ into a smaller positive exponential constant. Because $u(z) \leq C\|z\|^p$, this proves (14.2).

Given $\varepsilon > 0$, choose R large. If $d(z) > R$, then $z - b \in K$ and the uniform asymptotic for V gives $V(z - b)/u(z - b)$ close to one. Also

$$\frac{u(z - b)}{u(z)} \longrightarrow 1 \quad \text{uniformly as } d(z) \longrightarrow \infty. \quad (14.6)$$

To see the uniformity even when $\|z\|$ grows independently, use the explicit wedge function to obtain $\|\nabla \log u(w)\| \leq C/d(w)$. Along $w = z - tb$, $0 \leq t \leq 1$, the boundary distance is at least $d(z) - \|b\|$; integrate the logarithmic derivative along this bounded segment. Consequently, for sufficiently large R ,

$$V_b(z) \geq (1 - \varepsilon)u(z) \quad (d(z) > R). \quad (14.7)$$

Combining this inequality with (13.6) and (14.2) gives $\liminf_m P_0^m V_b(x) \geq (1 - \varepsilon)V(x)$. By (13.10), the corresponding upper bound is $P_0^m V_b(x) \leq V(x)$. Let $\varepsilon \downarrow 0$. $\square$

The argument needs only the ordinary iid harmonic asymptotic and survival bound. It does not assume a conditioned invariance principle for the valid-cycle kernel.

15 Harmonic and endpoint limits for valid cycles

Theorem 15.1. *The limit*

$$\mathcal{H}(x) = \lim_{m \rightarrow \infty} D^m V(x) \quad (15.1)$$

exists, is finite, is D -harmonic, and is positive at every starting point that can reach a point with $V_b > 0$ by a valid finite cycle path. Moreover,

$$n^{p/2} D^n \mathbf{1}(x) \longrightarrow \beta \mathcal{H}(x), \quad (15.2)$$

$$n^{p/2} \mathbb{E}_x[f(Z_n/\sqrt{n}); \text{the first } n \text{ cycles are valid}] \longrightarrow \beta \mathcal{H}(x) \nu(f) \quad (15.3)$$

for every bounded continuous nonnegative f .

Proof. The inequality $DV \leq P_0 V = V$ makes $D^m V$ nonincreasing and nonnegative, proving existence and finiteness. Since $DV_b \geq P_b V_b = V_b$ and $V \geq V_b$, one has $D^m V \geq V_b$. Dominated convergence against $D(x, dz)$, using $DV(x) \leq V(x) < \infty$, gives $D\mathcal{H} = \mathcal{H}$. Thus $\mathcal{H} \geq V_b$, and harmonicity propagates strict positivity backwards along every valid finite path to a point where $V_b > 0$.

Fix m and write $n = m + r$. Kernel order gives

$$D^m P_b^r \mathbf{1}(x) \leq D^n \mathbf{1}(x) \leq D^m P_0^r \mathbf{1}(x). \quad (15.4)$$

Multiply by $n^{p/2}$ and let $n \rightarrow \infty$ with m fixed. The ordinary survival theorem and its uniform polynomial upper bound permit dominated convergence against $D^m(x, dz)$. Indeed, D^m is bounded by the fixed-time iid displacement kernel, so this measure has all polynomial moments. It follows that

$$\beta D^m V_b(x) \leq \liminf_n n^{p/2} D^n \mathbf{1}(x) \leq \limsup_n n^{p/2} D^n \mathbf{1}(x) \leq \beta D^m V(x). \quad (15.5)$$

The width is at most

$$\beta D^m (V - V_b)(x) \leq \beta P_0^m (V - V_b)(x) \longrightarrow 0 \quad (15.6)$$

by Lemma 14.1. The upper endpoint tends to $\beta \mathcal{H}(x)$, proving (15.2).

For (15.3), insert the nonnegative terminal test function $f(z/\sqrt{n})$ in (15.4). For each fixed intermediate starting point, the ordinary endpoint theorem gives the same factor $\nu(f)$ in the inner and outer kernels: the shift is fixed and $r/n \rightarrow 1$. Domination follows from $\|f\|_\infty$ times the same survival bound. Thus the analogue of (15.5) has both endpoints multiplied by $\nu(f)$, and the same vanishing-width argument proves the result. Taking $f = 1$ gives convergence of total mass; together the statements give the normalized weak conditional endpoint limit. $\square$

16 Exact counted-time constants

Let R_n be survival of the original counted-step chain through time n . For an initial parity i , use its complete-cycle kernel D_i and the corresponding harmonic function $\mathcal{H}_i$; forward and reverse kernels may have different harmonic functions. Define

$$A(a) = 2^{p/2} \beta \mathcal{H}_{i(a)}(T(a + c_M)), \quad A^*(b) = 2^{p/2} \beta^* \mathcal{H}_{i(b)}^*(T(b + c_M)). \quad (16.1)$$

Here $i(a)$ and $i(b)$ denote the respective parities of the two endpoints. The starred harmonic normalization uses the same wedge function (13.3); writing β^* separately allows the ordinary reverse iid theorem to be applied without identifying its constant in advance. Its endpoint law is the same ν , since the covariance and wedge agree.

Theorem 16.1. *At every relevant forward and reverse starting point,*

$$\mathbb{P}_a(R_n) \sim A(a)n^{-p/2}, \quad \mathcal{L}_a(X_n/\sqrt{n} \mid R_n) \Longrightarrow \nu_\Sigma, \quad (16.2)$$

with the analogous reversed statement and amplitude $A^(b)$. The probability law ν_Σ is the endpoint law at time one of the Brownian meander in C with covariance Σ .*

Proof. Put

$$k_- = \lfloor n/2 - n^{3/4} \rfloor, \quad k_+ = \lceil n/2 + n^{3/4} \rceil. \quad (16.3)$$

Chernoff bounds for the iid durations, with mean two, show that outside an event of probability $O(e^{-c\sqrt{n}})$,

$$n - 4n^{3/4} \leq T_{k_-} \leq n \leq T_{k_+} \leq n + 4n^{3/4}. \quad (16.4)$$

The corrected-martingale exponential maximal inequality from the Foundation, applied on the deterministic interval beginning at $\lfloor n - 4n^{3/4} \rfloor$ and of length $O(n^{3/4})$, gives total positional oscillation at most $\delta\sqrt{n}$ outside an event of probability $O(e^{-c_\delta n^{1/4}})$. The bounded corrector contributes only $O(1)$. Both estimates are unconditional; no independence between duration and displacement and no conditioning on valid cycles is used.

For a continuity set B of the limiting endpoint law, let $B^{-\delta}$ and $B^{+\delta}$ be its inner and outer δ neighborhoods. The two complete-cycle events give

$$\begin{aligned} & \mathbb{P}\{D\text{-survival through } k_+, X_{T_{k_+}}/\sqrt{n} \in B^{-\delta}\} - \varepsilon_n \\ & \leq \mathbb{P}_a(R_n, X_n/\sqrt{n} \in B) \\ & \leq \mathbb{P}\{D\text{-survival through } k_-, X_{T_{k_-}}/\sqrt{n} \in B^{+\delta}\} + \varepsilon_n, \end{aligned} \quad (16.5)$$

where ε_n is exponentially small compared with $n^{-p/2}$. First choose δ so the neighborhood boundaries are continuity sets; then let $\delta \downarrow 0$. Since $k_\pm/n \rightarrow 1/2$, Theorem 15.1 yields a common amplitude factor $2^{p/2} \beta \mathcal{H}_i$. The spatial transformation is

$$\frac{1}{\sqrt{2}} T^{-1} = (\Sigma)^{1/2}, \quad (16.6)$$

so the pushed-forward endpoint law is precisely ν_Σ . The fixed domain translation vanishes under scaling. Using $B = \mathbb{R}^2$ gives the survival equivalent, and general continuity sets give the conditional weak limit. $\square$

16.1 Positivity at the enumeration endpoints

The valid entrances to the even inner cone were exhibited in Foundation Section 4.1. For completeness, a complete odd-return path for the reversed S chain from its odd endpoint is

$$(1, 1) \longrightarrow (0, 2) \longrightarrow (2, 2) \longrightarrow (4, 2) \longrightarrow (3, 3). \quad (16.7)$$

The first and last steps are reversed southeast steps. Each middle aggregate of displacement $(2, 0)$ reverses a southeast step followed by the face step $(-3, 1)$. Its internal physical points remain nonnegative. The final odd point lies in the inner domain; the middle internal states are even, so this is a complete first return to odd parity.

For reversed P from $(2i, 0)$, $i \geq 2$, two reversed southeast steps reach $(2i - 2, 2)$. For $i = 1$, the valid complete even-return path is

$$(2, 0) \longrightarrow (1, 1) \longrightarrow (3, 1) \longrightarrow (2, 2), \quad (16.8)$$

whose middle step is the reversed odd-to-odd face displacement $(2, 0)$. From these points the cycle-support accessibility constructions reach arbitrarily deep points where $V_b > 0$. Thus every endpoint in the P sum has a positive reversed amplitude, and all the fixed endpoints in (12.13) have positive amplitudes as well.

17 Uniform interior killed local convergence

Let $g_t^\Sigma(x, y)$ be the unrestricted Brownian transition density with covariance Σ , and let $k_t^{C, \Sigma}(x, y)$ be its density killed on leaving the open quadrant C . The distinction between the physical quadrant C and the whitened wedge $K = TC$ is maintained throughout this section.

Lemma 17.1. *Fix $t > 0$ and integers m_n with $m_n/n \rightarrow t$. For compact sets $E, F \subset C$,*

$$\sup_{a, b \in L: a/\sqrt{n} \in E, b/\sqrt{n} \in F} \left| nq_{m_n}(a, b) - 2k_t^{C, \Sigma}(a/\sqrt{n}, b/\sqrt{n}) \right| \longrightarrow 0. \quad (17.1)$$

Proof. We give the exit-decomposition proof. First, the unrestricted counted process satisfies a functional central limit theorem with covariance Σ . This follows from the bounded-corrector martingale: its predictable quadratic variation, divided by n , converges to $t\Sigma$ uniformly on compact time intervals by ergodicity of the finite internal chain, while the exponential moment implies the martingale Lindeberg condition. The bounded corrector disappears at scale $\sqrt{n}$. Equivalently, one may use the iid cycle functional limit theorem and the renewal law $T_k/k \rightarrow 2$; the dependent clock still converges to a deterministic time change. Exponential tails bound both maximum cycle length and maximum individual jump over $O(n)$ steps by $O(\log n)$ in probability, leaving an unfinished-cycle error $O(\log^2 n) = o(\sqrt{n})$.

It is enough first to describe $m_n = n$. The free exit decomposition is

$$q_n(a, b) = \mathbb{P}_a(X_n = b) - \mathbb{E}_a[\mathbb{P}_{X_\tau}(X_{n-\tau} = b); \tau \leq n], \quad (17.2)$$

where τ is the first exit from the discrete model domain. Fix $\varepsilon > 0$ and restrict the expectation to $\tau \leq (1 - \varepsilon)n$. The unrestricted functional limit theorem gives convergence of the rescaled exit

time and position on this truncated event. The continuous-mapping assertion uses familiar path properties of nondegenerate Brownian motion in a quadrant: it crosses a regular face immediately upon first hitting it, has zero probability of exit at the vertex, and its exit time has no atom at a fixed positive truncation time. The maximum jump divided by $\sqrt{n}$ tends to zero, so the discrete overshoot vanishes. The fixed displacement of the S boundary also vanishes.

On a compact set of rescaled exit positions, the free local theorem (12.10) applies uniformly, with remaining durations in $[\varepsilon n, n]$. After multiplication by n , its limit is twice the appropriate free Brownian density at the remaining time. Uniformity for the varying time ratios follows by compactness of $[\varepsilon, 1]$ and the local theorem, or by taking convergent subsequences of these ratios. Outside a compact set of exit positions, (12.9) bounds the rescaled integrand by C/ε , and functional-limit tightness of the path maximum removes that tail. Thus the truncated expectation converges to the corresponding Brownian exit integral.

It remains to control late exits. Reverse every path that ends at b and exits after time $(1 - \varepsilon)n$. Its reversal visits the complement within its first εn steps. Strong Markov at the first reversed exit, followed by the global free upper bound for at least $(1 - \varepsilon)n$ remaining steps, gives

$$\mathbb{P}_a(X_n = b, (1 - \varepsilon)n < \tau \leq n) \leq \frac{C}{n} \mathbb{P}_b^*(\tau^* \leq \varepsilon n). \quad (17.3)$$

If $b/\sqrt{n} \in F$ and F has boundary distance at least $\delta > 0$, the reversed corrected-martingale exponential inequality gives, for each fixed ε and sufficiently large n ,

$$\mathbb{P}_b^*(\tau^* \leq \varepsilon n) \leq C \exp[-c\delta^2/\varepsilon]. \quad (17.4)$$

Indeed, an exit requires a displacement of at least $\delta\sqrt{n} - O(1)$, and the optimized moment-generating parameter is of order $1/(\varepsilon\sqrt{n})$, within its allowed range for large n . Hence the late-exit mass times n tends uniformly to zero as $\varepsilon \downarrow 0$. The Brownian analogue removes the truncation in the limiting integral.

Subtracting the exit integral from the limiting free density in (17.2) is the Brownian Hunt formula and gives $2k_1^{C,\Sigma}$. The argument with $m_n/n \rightarrow t$ uses the identical truncation, with remaining time ratios bounded away from zero; it gives $2k_t^{C,\Sigma}$. Finally, compact uniformity follows by the subsequence principle: any sequence of endpoint pairs in $E \times F$ has a convergent subsequence, and a further subsequence fixes the finite parity pair. All the bounds above are uniform on those compact sets, and the limiting heat kernel is continuous there. $\square$

Only an unrestricted functional limit theorem and the already verified free local theorem enter this proof. No cone-killed local theorem for the two-state chain is assumed.

18 Fixed-thirds convergence and positive bridge amplitudes

Fix relevant endpoints a, b , put $m = \lfloor n/3 \rfloor$ and $\ell = n - 2m$, and define finite measures

$$\mu_{n,a} = n^{p/2} \sum_u q_m(a, u) \delta_{u/\sqrt{n}}, \quad \mu_{n,b}^* = n^{p/2} \sum_v q_m^*(b, v) \delta_{v/\sqrt{n}}. \quad (18.1)$$

Let $\nu_{1/3,\Sigma}$ be the image of ν_Σ under $x \mapsto x/\sqrt{3}$. By Theorem 16.1, including convergence of total mass,

$$\mu_{n,a} \Longrightarrow 3^{p/2} A(a) \nu_{1/3,\Sigma}, \quad \mu_{n,b}^* \Longrightarrow 3^{p/2} A^*(b) \nu_{1/3,\Sigma}. \quad (18.2)$$

For this model define the positive continuum constant

$$\mathcal{B}_\Sigma = 2 \cdot 3^p \iint_{C \times C} k_{1/3}^{C, \Sigma}(x, y) \nu_{1/3, \Sigma}(dx) \nu_{1/3, \Sigma}(dy). \quad (18.3)$$

It is finite by the unrestricted Gaussian bound and strictly positive because the killed heat kernel and the meander density are positive in the interior.

Theorem 18.1 (Fixed-endpoint amplitude). *For each relevant pair of fixed endpoints,*

$$q_n(a, b) \sim \mathcal{C}(a, b) n^{-p-1}, \quad \mathcal{C}(a, b) = \mathcal{B}_\Sigma A(a) A^*(b) > 0. \quad (18.4)$$

Proof. The exact semigroup identity and reversal give

$$n^{p+1} q_n(a, b) = \iint n q_\ell(u, v) \mu_{n,a}(d(u/\sqrt{n})) \mu_{n,b}^*(d(v/\sqrt{n})). \quad (18.5)$$

On compact interior sets, Lemma 17.1 makes the kernel converge uniformly to $2k_{1/3}^{C, \Sigma}$. The global bound $nq_\ell \leq C$ holds everywhere by (12.9). The finite measures in (18.2) are tight and have convergent total masses. Their limits put no mass on the boundary and none at infinity. Therefore compact interior sets, chosen with limiting boundary measure zero, leave arbitrarily small mass in the complement, uniformly for large n . The contribution to (18.5) from either complementary coordinate is bounded by that small mass times C and the bounded total mass of the other factor. On the compact product, weak product-measure convergence and uniform kernel convergence give the integral in (18.3), multiplied by $A(a)A^*(b)$. This proves (18.4). $\square$

The argument uses a fixed positive fraction of the time in all three pieces. It needs neither a marked-clock local theorem in three dimensions nor a shrinking-time gluing limit.

[**write, 2026-10-05**] This is the fixed-endpoint killed local limit that Report 112 (Part III, Section 30) names as the stronger result its method cannot give, here proved for the two models.

18.1 Evaluation of the universal Brownian factor

The positive integral (18.3) is the primary definition of the bridge factor. Its universal part can also be evaluated explicitly, while the discrete harmonic values remain unevaluated. All Brownian densities below use generator $\frac{1}{2}\Sigma : \nabla^2$ and physical planar Lebesgue measure. Write

$$h(x) = u(\Sigma^{-1/2}x), \quad C_B = \frac{2^{1-p}}{\theta \Gamma(p+1) \sqrt{\det \Sigma}}, \quad \beta_B = \frac{2^{1-p/2} \Gamma(p/2)}{\pi \Gamma(p)}. \quad (18.6)$$

The covariance-identity wedge kernel, obtained from the spectral/Bessel formula of Bănelos–Smits [10, Lemma 1, equation (2.2)], is

$$k_t^K((r, \varphi), (s, \psi)) = \frac{2}{\theta t} e^{-(r^2+s^2)/(2t)} \sum_{j \geq 1} I_{jp}(rs/t) \sin(jp\varphi) \sin(jp\psi). \quad (18.7)$$

Its leading term uses $I_p(z) \sim (z/2)^p / \Gamma(p+1)$. After the physical coordinate change, the small-start entrance density, divided by the starting harmonic value, is

$$g_t(y) = C_B t^{-p-1} h(y) e^{-y^T \Sigma^{-1} y / (2t)}. \quad (18.8)$$

The angular and radial integrals are

$$\int_0^\theta \sin(p\varphi) d\varphi = \frac{2}{p}, \quad \int_0^\infty r^{p+1} e^{-r^2/(2t)} dr = 2^{p/2} t^{p/2+1} \Gamma(p/2 + 1). \quad (18.9)$$

Since $\theta p = \pi$, integration of (18.8) gives $\int_C g_t(y) dy = \beta_B t^{-p/2}$. Thus

$$\nu_{t,\Sigma}(dy) = \frac{g_t(y) dy}{\beta_B t^{-p/2}}. \quad (18.10)$$

With the harmonic normalization (13.3), the ordinary iid survival constant is this Brownian constant, so $\beta = \beta^* = \beta_B$.

Define the physical harmonic functions, with the parity of their argument selecting the relevant cycle law, by

$$U_i(a) = 2^{p/2} \mathcal{H}_i((2\Sigma)^{-1/2}(a + c_M)), \quad U_i^*(a) = 2^{p/2} \mathcal{H}_i^*((2\Sigma)^{-1/2}(a + c_M)). \quad (18.11)$$

Then $A = \beta_B U$ and $A^* = \beta_B U^*$. They are harmonic for the respective killed counted-time kernels: apply the one-step survival recursion, multiply by $n^{p/2}$, and pass to the limit using (12.8) and the exponential one-step moment as an integrable majorant. The bounds $V_b \leq \mathcal{H} \leq V$ and the fixed-shift comparison give $U(a) \sim h(a)$ as the boundary distance tends to infinity. No uniqueness assertion is required.

The entrance densities satisfy $g_s K_t = g_{s+t}$, by the killed semigroup and its small-start limit. This also follows from (18.7), angular orthogonality and the termwise Bessel identity

$$\int_0^\infty r^{p+1} e^{-ar^2} I_p(br) dr = \frac{b^p e^{b^2/(4a)}}{(2a)^{p+1}}, \quad a > 0. \quad (18.12)$$

For $s = 1/3$, each limiting measure in (18.2) is $U(a)g_s(x) dx$. The limiting bridge integral therefore reduces to

$$2U(a)U^*(b) \int_C g_{2s}(y)g_s(y) dy. \quad (18.13)$$

A polar integration, using $\int_0^\theta \sin^2(p\varphi) d\varphi = \theta/2$, gives

$$\begin{aligned} \int_C g_{2s}(y)g_s(y) dy &= C_B^2 \sqrt{\det \Sigma} \frac{\theta \Gamma(p+1)}{4} (2s)^{-p-1} s^{-p-1} (4s/3)^{p+1} \\ &= C_B (3s)^{-p-1} = C_B. \end{aligned} \quad (18.14)$$

Consequently the equivalent explicit form of (18.4) is

$$\begin{aligned} \mathcal{C}(a, b) &= 2C_B U(a)U^*(b) \\ &= \frac{4}{\theta \Gamma(p+1) \sqrt{\det \Sigma}} \mathcal{H}_i(T(a + c_M)) \mathcal{H}_j^*(T(b + c_M)). \end{aligned} \quad (18.15)$$

The factor two in the first line is the discrete lattice covolume; it is not part of the Brownian density. Here i, j are the initial and terminal parities. In particular, the S bridge uses the even forward and odd reversed harmonic functions. The determinants are

$$\det \Sigma_P = 7, \quad \det \Sigma_S = 80/9. \quad (18.16)$$

As normalization checks, for identity covariance and $p = 1$ the constants are $C_B = 1/\pi$, $\beta_B = \sqrt{2/\pi}$; for $p = 2$, they are $C_B = 1/(2\pi)$, $\beta_B = 1/\pi$, with $u(x, y) = 2xy$. These formulas evaluate only the universal Brownian factors. The discrete harmonic values, and the weighted endpoint sum for P , still determine the counting amplitudes.

19 Enumeration amplitudes and the rigid sequence

Attach subscripts P, S to the constants for the two models.

19.1 Corner polyhedra

By (12.12), Theorem 18.1, and the uniform bound (12.11), dominated convergence in the geometrically weighted endpoint sum gives

$$\kappa_P = \sum_{i \geq 1} 3^{-i} \mathcal{C}_P((0, 0), (2i, 0)) = \mathcal{B}_{\Sigma_P} A_P((0, 0)) \sum_{i \geq 1} 3^{-i} A_P^*((2i, 0)). \quad (19.1)$$

The dominating series $\sum_{i \geq 1} 3^{-i} (1 + i)^{p_P}$ converges. Positivity follows already from the single endpoint $(4, 0)$. This proves (11.2).

19.2 Schnyder labelings

The fixed-endpoint identity (12.13) gives

$$s'_n \sim \kappa'_S \mu_S^n n^{-\alpha_S}, \quad \kappa'_S = \frac{\mu_S}{4} \mathcal{C}_S((0, 2), (1, 1)). \quad (19.2)$$

Its positive finite combination then gives

$$\kappa_S = (1 + \mu_S^{-1})^2 \kappa'_S = \frac{361}{192} \mathcal{B}_{\Sigma_S} A_S((0, 2)) A_S^*((1, 1)). \quad (19.3)$$

Here $\mu_S/4 = 4/3$ and $(1 + \mu_S^{-1})^2 = (19/16)^2$. This proves (11.3).

19.3 A dominated signed convolution

Write

$$(1 + z)^{-3} = \sum_{j \geq 0} b_j z^j, \quad b_j = (-1)^j \binom{j+2}{2}. \quad (19.4)$$

From (12.14), $\tilde{s}_n$ is the signed convolution of s_n with b_j , minus the coefficient of $(3z^2 + z^3)/(1 + z)^3$. Enlarge a constant so that

$$0 \leq s_m \leq C \mu_S^m (m + 1)^{-\alpha_S} \quad (m \geq 0). \quad (19.5)$$

For $0 \leq j \leq n$,

$$\frac{n}{n - j + 1} \leq j + 1, \quad (19.6)$$

so the absolute normalized convolution summand is bounded by

$$n^{\alpha_S} \mu_S^{-n} |b_j| s_{n-j} \leq C |b_j| \mu_S^{-j} (j + 1)^{\alpha_S}. \quad (19.7)$$

The right side is summable over j , independently of n . For each fixed j , the summand has signed limit $\kappa_S b_j \mu_S^{-j}$. Dominated convergence therefore gives

$$\lim_{n \rightarrow \infty} n^{\alpha_S} \mu_S^{-n} \sum_{j=0}^n b_j s_{n-j} = \kappa_S \sum_{j \geq 0} b_j \mu_S^{-j} = \frac{\kappa_S}{(1 + \mu_S^{-1})^3}. \quad (19.8)$$

The polynomial correction divided by $(1 + z)^3$ has coefficients $O(n^2)$, so it is exponentially negligible under this normalization. Since $(1 + \mu_S^{-1})^{-3} = (16/19)^3$, (11.4) follows. This completes the proof of Theorem 11.1.

20 Sharper first-threshold inversion

The following conclusion applies to each of the three counting sequences, with its rate μ , exponent α and amplitude γ from Theorem 11.1.

Theorem 20.1. *Suppose $a_n \sim \gamma \mu^n n^{-\alpha}$ with $\gamma > 0$, $\mu > 1$ and $\alpha > 0$, and define $N(Y) = \min\{n \geq 0 : a_n \geq Y\}$. Put $\lambda = \log \mu$, $T_Y = \log Y$, and*

$$r(Y) = -\frac{\alpha}{\lambda} W_{-1} \left(-\frac{\lambda}{\alpha} (\gamma/Y)^{1/\alpha} \right). \quad (20.1)$$

Then

$$r(Y) = \frac{T_Y + \alpha \log T_Y - \alpha \log \lambda - \log \gamma}{\lambda} + o(1). \quad (20.2)$$

There is a positive function $\varepsilon_Y \rightarrow 0$ such that, for all sufficiently large Y ,

$$\lceil r(Y) - \varepsilon_Y \rceil \leq N(Y) \leq \lceil r(Y) + \varepsilon_Y \rceil. \quad (20.3)$$

No separate monotonicity assumption on a_n is needed.

Proof. The defining relation $W(z)e^{W(z)} = z$ gives, on the negative real branch W_{-1} ,

$$\gamma e^{\lambda r(Y)} r(Y)^{-\alpha} = Y. \quad (20.4)$$

The branch selects the large positive solution tending to infinity; the other real branch selects the small solution. To obtain (20.2), first $r(Y) \sim T_Y/\lambda$ by taking logarithms in (20.4); substituting this into $\lambda r = T_Y + \alpha \log r - \log \gamma$ gives the stated expansion, with $o(1)$ remainder after one substitution.

Write $\log a_n = g(n) + e_n$ for all sufficiently large n , where

$$g(t) = \lambda t - \alpha \log t + \log \gamma, \quad e_n \rightarrow 0. \quad (20.5)$$

The coefficients are eventually positive by the full equivalent. For large t , $g'(t) \geq \lambda/2$. Put $\delta_Y = \sup_{n \geq r(Y)/2} |e_n|$, where the supremum is over integers in that range. Then $\delta_Y \rightarrow 0$. Choose $\varepsilon_Y > 2\delta_Y/\lambda$ with $\varepsilon_Y \rightarrow 0$, for instance $3\delta_Y/\lambda + 1/r(Y)$.

Every integer n with $r(Y)/2 \leq n < r(Y) - \varepsilon_Y$ satisfies $g(n) + e_n < g(r(Y)) = \log Y$. Conversely the integer $n = \lceil r(Y) + \varepsilon_Y \rceil$ satisfies $g(n) + e_n > \log Y$. Integers below $r(Y)/2$ remain below threshold for large Y by the exponential-polynomial upper bound and eventual increase of its comparison envelope; finitely many initial values cause no exception for large Y . This proves the two ceiling bounds. $\square$

The conclusion is deliberately a bracket. There is no unconditional assertion that $N(Y) = \lceil r(Y) \rceil$, because $r(Y)$ can approach an integer more closely than the unknown forward relative error permits. Nor do additional formal terms of the Lambert- W model provide a justified error expansion for the actual counting sequence without a quantitative remainder in its forward asymptotic. No numerical amplitude estimate is substituted into (20.1).

[write, 2026-10-02] Theorem 20.1 is an instance of the transseries volume's `p0:thm:lambert-core` ($a = \lambda$, $b = -\alpha$, branch W_{-1}), with (20.2) the first step of `p0:thm:lambert-centered` and the bracket (20.3) the separation situation of `p0:thm:staircase` (Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/). No novelty is claimed for the inversion mechanics. The generic staircase separation lemma is formalized in Lean as `Fabius.staircase_separation`; nothing about p_n , s_n or $\tilde{s}_n$ is formalized.

21 Verification, reproducibility and limits of the result

The accompanying package contains the addendum source and PDF, the unchanged 30-file Foundation package, portable build commands, exact symbolic checks of the new algebraic amplitude factors and inverse identities, source and mathematical-review receipts, and checksums. The supplied integrity program verifies every included Foundation file against its original manifest before any reproduction command is run.

[write, 2026-10-02] In this repository the addendum PDF, the embedded Foundation copy, the integrity receipt `results/foundation-integrity.json` (a hash ledger of that copy) and the checksum ledgers are not shipped, so `code/46-amp-verify_foundation.py` and `code/46-amp-run_all.py` cannot run from the report directory. The report README says how to rerun them from the archive in the arrival commit, and how to run the new algebra check `code/46-amp-verify_addendum.py` alone.

Finite computations verify algebraic identities, normalization factors and indexing. They do not establish Lemma 14.1, the valid-cycle harmonic sandwich, the functional-limit and exit-decomposition argument, or the limiting interchange in the semigroup identity; those are the mathematical proofs given above. The positive amplitude formulas (19.1) and (19.3) characterize the constants through killed harmonic functions and Brownian heat kernels. They do not certify numerical amplitude digits, a rate of convergence, or any unproved correction term.

Part III

A log-scale cone theorem for state-modulated walks (source 47)

Logarithmic asymptotics for polyhedral orientations and Schnyder labelings

Research report 112, 2 October 2026

Abstract of source 47 (bundle Report 112), as delivered. This report treats polyhedral orientations counted by OEIS A377922 and 3-connected Schnyder labelings counted by A377920. It gives a finite-phase cone argument with exponentially decaying jump tails, then verifies the exact walk models, counting time, boundary conditions, covariance and endpoint normalizations needed for the two applications. The conclusions are

$$p_n = (9/2)^n n^{-\alpha_P + o(1)}, \quad s_n = (16/3)^n n^{-\alpha_S + o(1)},$$

where $\alpha_P = 1 + \pi / \arccos(9/16)$ and $\alpha_S = 1 + \pi / \arccos(22/27)$. The growth constants were already proved by Fusy, Narmanli and Schaeffer in 2023. The advance established here is the logarithmic polynomial power, with non-D-finiteness and two-term first-threshold inverses. An explicit six-face sleeve injection transfers the all-index logarithmic power, non-D-finiteness and inverse expansion to A377921. Neither a leading amplitude nor a full multiplicative equivalent is obtained.

[**write, 2026-10-05**] Report 112 was written without knowledge of Parts I and II, which prove more about the three sequences: Θ bounds (Theorem 1.1), full equivalents with positive amplitudes (Theorem 11.1) and inverses with $O(1)$ error and a next constant. Its sentence “the advance established here is the logarithmic polynomial power” is therefore true only of Report 112 alone. What Part III adds to the report is listed in Section 22: chiefly a general cone theorem, Theorem 24.1, and a six-face sleeve injection, Proposition 29.1.

22 Guide to Part III

This section was written when bundle Report 112 was added as Part III (5 October 2026). Everything after it, up to Appendix A, is Report 112 as delivered, apart from labels and the changes listed below, with dated notes marked [**write, 2026-10-05**]. In Part III, “this report” and “the source” are Report 112’s words: the first means Report 112, the second the paper of Fusy, Narmanli and Schaeffer [1].

22.1 What is new, and what is a second route

Part III proves no asymptotic statement about A377922, A377920 or A377921 that Parts I–II do not already prove in a stronger form. Its statements about the three sequences are a second route to Part I, by a different method and with weaker conclusions (logarithmic scale, $n^{o(1)}$ instead of Θ or $\sim$); Table 2 maps each of them to its counterpart. New to the report are:

- (i) *Theorem 24.1*, a cone theorem for a Markov-additive chain on $\mathbb{Z}^2 \times E$ with a finite internal state (phase) set E and unbounded steps with a uniform exponential moment, killed on leaving the closed quadrant: under hypotheses (H1)–(H5), $K_n((0, a), (0, b)) = n^{-1-\pi/\theta+o(1)}$ at every integer time, and under (H1)–(H4) the survival bound $\mathbb{P}(\tau > n) \leq C_\epsilon n^{-\nu/2+\epsilon}$. Part I states that it “does not posit a new general cone theorem for arbitrary Markov-modulated walks”; Theorem 24.1 is such a theorem, at the logarithmic scale only. Lemma 24.2, its exact-time interior bridge bound, is the general form of Lemma 8.1.

- (ii) *Proposition 29.1*, an explicit injection from Schnyder labelings of size n into those of size $n + 6$ whose outer white vertices are non-isolated (a six-face annulus around the outer hexagon), hence $s_{n-6} \leq b_n \leq s_n$ for $n \geq 8$. Neither Parts I–II nor [1] as cited contain it.
- (iii) *A non-D-finiteness criterion with a weaker hypothesis* (Section 27 and Remark 27.1): non-negative integer coefficients with $a_n = \Gamma^n n^{-\alpha+o(1)}$, $\Gamma > 1$ and α irrational already exclude D-finiteness. Lemma 9.2 needs $a_n = \Theta(\mu^n n^{-\alpha})$.
- (iv) *The fixed-endpoint sandwich* $e_{n-1} \leq p_n \leq e_{n+1}$ of (25.6), with e_L the number of legal corner paths of length L from $(0, 0)$ to $(1, 1)$, by two explicit injections. It replaces the tilted endpoint sum (2.6) by one fixed endpoint of tilt one.
- (v) *An explicit exponential-moment certificate* for a whole Schnyder aggregate, (26.7): at $e^t = 6/5$ the moment of the total elementary length is $189/13$. Part I obtains the moment from analyticity of the return generating function (3.9). (The delivered checks also certify the corner quotient moment $524/405$ at the same point; see Section 32.)
- (vi) *Exact bounds* $4 < \alpha_P < 5$ and $6 < \alpha_S < 7$ (Section 27); Parts I–II give decimals only.
- (vii) *Three references* not cited in Parts I–II: Denisov–Zhang [4], Grama–Lauvergnat–Le Page [5] and Pham–Peigné–Son [6] (Section 30), together with Whitt’s martingale functional limit theorem [3], which Theorem 24.1 uses.

Also new are several derivations of known quantities: a probabilistic reconstruction of the corner kernel, direct second-moment computations of both covariances, and a second proof of the aperiodicity Lemma 7.1 with new seed paths. Report 112’s further questions (Section 31) are mostly answered by Part II; dated notes there say which.

22.2 Provenance

Part III is bundle Report 112. Its archive is `Geometric_OEIS_Sequences_Cone_Exponents_and_Inverses_Source.zip` (wrapper `report112/`, main file `report112.tex`, a 19-page PDF). It arrived in commit `60f54ea06` and was placed as source 47 of this report, manuscript 01 of cluster 103-A377922 of batch 103, in `9c995cefe`. Its author line and PDF metadata say “Research report 112”; it names no tool and no person. It is dated 2 October 2026 and names no ProveIt commit, path or search: there is no pin. It was written about eight hours after sources 45 and 46 (their literature snapshots are 01:22–01:39 UTC on 2 October; the archive was created at 09:26 UTC) in a different session, and it shows no knowledge of them. Its text carries no “prepared for private review” line and no personal data. Its programs, fixtures and recorded outputs are shipped byte-identical under the prefix `47-logcone-`; the manuscript, its PDF, its README and its checksum inventory `manifest.json` are not shipped (all in the arrival commit). The report README lists the files and gives a rerun route.

22.3 How Report 112 is printed

- Report 112 is printed in full, with its abstract. Its Section k is Section $k + 22$ here, and its Theorem, Lemma, Proposition or Corollary $k.m$ is $(k + 22).m$; so its Theorem 2.1 is Theorem 24.1 and its Proposition 7.1 is Proposition 29.1. Its equations, numbered consecutively as delivered, are numbered within sections here.
- Its statements about the three sequences are printed, not replaced by pointers, because they are where Theorem 24.1’s hypotheses are verified for the two models, and because the new

Table 2: Report 112 mapped to Parts I–II. “Second route” means a different proof of a conclusion that Parts I–II already contain, here with the strength stated.

Report 112	Here	Parts I–II	Class
Thm 1.1: p_n, s_n on the log scale	Thm 23.1	Θ : Thm 1.1, (1.3); $\sim$: Thm 11.1	second route, weaker
Thm 1.1, Cor 7.2: b_n on the log scale	(23.3), Cor 29.2	$\tilde{s}_n \sim (16/19)^3 \kappa_S \mu_S^n n^{-\alpha_S}$, (11.4)	second route, weaker; needs no amplitude
Thm 1.1: three GFs not D-finite	Thm 23.1, Secs 27, 29	Cor 9.4	same theorem; criterion new (iii)
Thm 1.1, Sec 6: inverse, $o(\log \log y)$	(23.4), Sec 28	$O(1)$: Lemma 9.5; next constant: Thm 20.1	weaker
Thm 2.1, general cone theorem	Thm 24.1	disclaimed (Part I, Sec. 1)	new (i)
Lemma 2.2, interior bridge	Lemma 24.2	Lemma 8.1 (two models)	general form
Secs 2.2, 2.4–2.5: FCLT, LLT, annular crossings, first-hit gluing	Secs 24.2, 24.4–24.6	Prop 7.2; Lemma 17.1 uses an FCLT; Part I glues by regeneration with a cycle buffer	general; crossing method new to the report
Quotient kernels $P(u, v)$, $G(u, v)$; corner reconstruction	(25.2), (26.5)	K_P, K_S : (2.4), (2.11)	same kernels in half-scale coordinates; new derivation
Σ_P, Σ_S , drifts, correctors, direct moments	(25.3), (26.6)	(3.7), (3.3)	same, divided by 4 (Remark 22.1)
(H4) aperiodicity, (H5) seeds	Secs 25, 26	Lemma 7.1; Sec. 4.1	second proof; different seeds
$e_{n-1} \leq p_n \leq e_{n+1}$	(25.6)	tilted sum (2.6)	new (iv)
Aggregates, shifted quadrant, endpoint factor 4	Sec. 26, (26.8)	Lemma 2.1, (2.13)	same, second proof
Moment 189/13	(26.7)	analyticity of (3.9)	new (v)
Irrationality; $4 < \alpha_P < 5$, $6 < \alpha_S < 7$	Sec. 27	Lemma 9.1; decimals	same proof; bounds new (vi)
Prop 7.1, sleeve injection	Prop 29.1	none	new (ii)
Sec 7.2: convolution relation	(29.4), (29.5)	(9.4)	same identity (Remark 29.4)
Sec 8: phase covariances V_c^P, V_c^S	Sec. 30	(3.5), (3.6) and the remark after (3.7)	same, divided by 4
Sec 8: literature	Sec. 30	[2] only	three references new (vii)
Sec 9: questions	Sec. 31	answered in part by Part II	dated notes

items (iv) and (v) sit inside them. The dated notes and Table 2 mark them as second routes.

- Labels carry the prefix `cps:log:`. Report 112's macros are mapped to this report's ($\mathbb{E}$, $\mathbb{R}$, $\mathbb{Z}$ by `\EE`, `\RR`, `\ZZ`); its title block, table of contents and running heads are omitted. No statement, proof, symbol or number was changed.
- Bibliography: Report 112's entries for Fusy–Narmanli–Schaeffer and Fischler–Rivoal are the same papers as [1] and [8]; its DW is [2]. Its entries for Whitt, Denisov–Zhang, Grama–Lauvergnat–Le Page and Pham–Peigné–Son are added. Its uncited OEIS entry (A377922, A377920, A377921, A377923) duplicates [11, 12], except for the excluded A377923, and is omitted.
- Report 112 cites [1, Proposition 19] for the shift relation (26.9), which Part I cites as Proposition 14 and Remark 9. Both are right: in arXiv:2202.09172v3 Remark 9 gives the refined relation by numbers of white and black vertices, and Proposition 19 the relation (2.8) for $n \geq 4$ (checked 5 October 2026). Report 112's form (29.4) of Remark 20 is the one printed there.

22.4 Notation

Report 112 works in *half-scale* (quotient) coordinates: a point of the parity lattice L is written $x = 2U + c$, $y = 2V + c$ for corner polyhedra (25.1) and $x = 2U + c$, $y = 2V + 2 - c$ for Schnyder labelings (26.3), with phase $c \in \{0, 1\}$ equal to the parity of y . Phase 0 is Parts I–II's even state e , phase 1 the odd state o . For corner polyhedra (U, V) are Part I's cell coordinates (a, b) of Section 7; for Schnyder labelings the cell coordinates are $(U, V + 1 - c)$, because Report 112 shifts the quadrant Q_S to $U, V \geq 0$. Time is the same as in Parts I–II: physical steps for corner polyhedra, aggregates for Schnyder labelings. The letters are kept; Table 3 lists the clashes.

Remark 22.1 (write, 2026-10-05: the factor 4). Part III's covariances are a quarter of Parts I–II's: $\Sigma_P^{\text{III}} = \Sigma_P/4$, $\Sigma_S^{\text{III}} = \Sigma_S/4$, and the phase-conditional covariances of Section 30 are $V_0^P = C_e^P/4$, $V_1^P = C_o^P/4$, $V_0^S = C_e^S/4$, $V_1^S = C_o^S/4$ (compare (25.3), (26.6) with (3.7), (3.5), (3.6)). The drifts and correctors correspond as

$$d_c = 2m_c + (P_{c,1-c})(1 - 2c)(1, \pm 1), \quad h_c^{\text{I}} = 2h_c - c(1, \pm 1) + \frac{1}{2}(1, \pm 1),$$

with the upper sign for corner polyhedra and the lower for Schnyder labelings, where h^{I} is Part I's corrector of (3.3). Correlations, the angle θ , $\nu = p = \pi/\theta$ and the exponents are the same.

Proof. Write the physical position as $X_n = 2S_n + \phi(C_n)$, with $\phi(c) = (c, c)$ for corner polyhedra and $\phi(c) = (c, 2 - c)$ for Schnyder labelings, and $S_n = (U_n, V_n)$. A step from phase c to phase d has physical displacement $2\Delta S + \phi(d) - \phi(c)$, and $\phi(d) - \phi(c) = (d - c)(1, \pm 1)$. Taking the conditional mean, $\mathbb{E}_c[d - c] = P_{c,1-c}(1 - 2c)$, which gives the first identity. If $Z_n = S_n + h(C_n)$ is the martingale of Section 24.2, then $2Z_n = X_n - \phi(C_n) + 2h(C_n)$, so $X_n + h^{\text{I}}(C_n)$ is a martingale for $h^{\text{I}} = 2h - \phi + k$ with any constant k ; Part I's normalization $h_o^{\text{I}} = -h_e^{\text{I}}$ fixes k , and since $\phi(c) = c(1, \pm 1) + (0, 1 \mp 1)$ and $h_1 = -h_0$, this gives the second identity. Its increments are twice the increments of Z , so every conditional and every averaged covariance of the physical corrected increment is four times the corresponding quarter-scale one. With the numbers: corner polyhedra, $m_0 = (-1/3, -1/3)$, $P_{01} = 5/6$, $d_e = (-2/3 + 5/6)(1, 1) = (1/6, 1/6)$, $h_0 = (-1/5, -1/5)$, $h_e^{\text{I}} = (-2/5 + 1/2)(1, 1) = (1/10, 1/10)$; Schnyder labelings, $m_0 = (-1/2, 3/8)$, $P_{01} = 7/8$, $d_e = (-1 + 7/8, 3/4 - 7/8) = (-1/8, -1/8)$, $h_0 = (-2/7, 3/14)$, $h_e^{\text{I}} = (-4/7 + 1/2, 3/7 - 1/2) = (-1/14, -1/14)$, as in (3.3). Scaling a covariance does not change a correlation, hence not θ . $\square$

Table 3: Letters of Part III that clash with Parts I–II. “Reading to avoid” is the tempting false identification.

Letter	Part III (source 47)	Parts I–II	Reading to avoid
Σ_P, Σ_S	covariance per unit time in half-scale coordinates	the same per physical unit	they are equal; Part III’s is a quarter (Remark 22.1)
V_c^P, V_c^S	phase-conditional covariances (Section 30)	C_i^P, C_i^S , four times larger; Part II’s V is a harmonic function	
m, h	quarter-scale drift and corrector, $(I - P)h = m$	d, h physical	$h_0 = h_e^I$
Γ_P, Γ_S	growth rates 9/2, 16/3	μ_P, μ_S ; Part II’s Γ is the Gamma function	
ν	π/θ	p ; Part II’s ν is the meander endpoint law	ν a measure
c, d, a, b	phases 0, 1 (0 = e, 1 = o); a, b endpoint phases in Thm 24.1	a, b points	
e_L	legal corner paths $(0, 0) \rightarrow (1, 1)$ of length L	e the even state	
P	phase matrix P_{cd} ; corner kernel $P(u, v)$	P the model; $P = K(1, 1)$	$P(u, v)$ is K_P in half-scale variables, not $K_P(u, v)$
$p_{cd}(v)$	transition probabilities	$p = \pi/\theta$	
D	$(1 - z/u)(1 - zv)$, Part I’s D at $u = x^2, v = y^2$; a constant in $L_n = D \log(n + 2)$	Part II: valid-cycle kernel	
A	aggregate matrix ($= M_S$); $A(t) = S(z)$; tail constant in (24.8)	$A(z)$ in Lemma 9.2; Part II amplitudes $A(a)$	
F	face matrix ($= W$); Fourier matrix; $F(z)$ a generating function	Fourier matrix $F(t)$ of cells	
J	SE matrix ($= E$); angular interval in (H5); a stage index	the state J_t	
W	whitening $\Sigma^{-1/2}$	Part I: face matrix; Part II whitens with $T = (2\Sigma)^{-1/2}$	
T	corrected increment; time constant $T(q)$	Part I: $T(z)$; Part II: whitening	
G, C	Schnyder quarter-scale kernel and its scalar $C(u, v)$	K_S ; Part II’s C the open quadrant	
K_n	killed kernel	q_n ; K_P, K_S step matrices	
$b_n, B(t)$	A377921 and its series	$\tilde{s}_n, T(z) = \tilde{S}(z)$; Part II’s b_j, b_P, b_S	
H	$H(t) = F(Rt)$; Dirichlet solution H_f	Schnyder denominator; Part II’s $\mathcal{H}$	
κ	exponential tail rate	Part II amplitudes κ_P, κ_S	
E	finite phase set	SE matrix $E(x, y)$	
L_n, L, R	jump cutoff; face sums; $R = 1/\Gamma$, radii	lattice L ; return function R_i	
$N(y)$	first threshold at level y	$N(Y)$	

22.5 Reports that use the same method

Report 112 is one of three bundle reports of 2 October 2026 (archives created 08:51, 09:26 and 14:23 UTC) that run one argument: a Perron tilt to a centred walk with a finite phase, a bounded Poisson corrector and the martingale functional limit theorem, a Fourier-matrix local limit theorem, an exact-time interior bridge, Brownian wedge harmonic measure over geometric annuli, two first-hit prefixes glued at the exact time, then non-D-finiteness by a positive derivative and the G -function theorem, and an $o(\log \log y)$ inverse. Report 111 proves it for one-sided rectangulations, A348351, with bounded steps; it is the report **a348351-one-sided-rectangulations** in the same collection directory as this one, placed in 9c995cfe. Report 123 proves it for the inversion-sequence class A279571; it arrived in 60f54ea06 and has not been placed (5 October 2026). Reports 111 and 123 argue model by model; Report 112 alone states the argument as a theorem, Theorem 24.1. Report 111 proves $\log a(n) = n \log \Gamma - \alpha \log n + o(\log n)$ for A348351, with $\Gamma = (7 + \sqrt{17})/2$, for a four-colour quadrant walk with bounded steps; it does not cite Report 112, nor Report 112 it. That report's Remark 1.2 (**osr:rem:cone**) checks that its walk satisfies (H1)–(H5) of Theorem 24.1 at the phases (c, W) , $c \in \{B, R, G\}$ (primitive colour matrix, bounded steps, zero drift with positive definite covariance, its own aperiodicity section, explicit seeds from both endpoints), so its exponent is also an instance of Theorem 24.1; Report 111 is the earlier, model-specific proof (bundle index 08:51 against 09:26 UTC). Its non-D-finiteness argument is the one of Section 27, and its Remark 10.1 (**osr:rem:criterion**) states the criterion in general, as Remark 27.1 does here. Its irrationality certificate differs from Lemma 9.1: there $2 \cos \theta = (-29 + 7\sqrt{17})/2$ is an algebraic integer, and the certificate is its conjugate, which lies below -2 . No text is shared, and the reports stay separate.

23 Results and their precise scope

Write $p_n = A377922(n)$ and $s_n = A377920(n)$, both with index starting at zero. The former counts polyhedral orientations with n inner vertices; the latter counts all 3-connected Schnyder labelings with n inner faces in the source's $(6, 4)$ -dissection model. The established bijections and conventions are those of [1], Propositions 13–14, 19 and 23. These definitions matter: A377923 counts corner-polyhedron *graphs*, which are different objects and are excluded here.

For a positive sequence a_n , the notation

$$a_n = \Gamma^n n^{-\alpha+o(1)}$$

means

$$\lim_{n \rightarrow \infty} \frac{\log(a_n/\Gamma^n)}{\log n} = -\alpha.$$

It does not mean that $a_n/(\Gamma^n n^{-\alpha})$ converges.

Theorem 23.1 (The three counting sequences). *Let $b_n = A377921(n)$ and*

$$\Gamma_P = \frac{9}{2}, \quad \alpha_P = 1 + \frac{\pi}{\arccos(9/16)}, \quad \Gamma_S = \frac{16}{3}, \quad \alpha_S = 1 + \frac{\pi}{\arccos(22/27)}.$$

Then

$$p_n = \Gamma_P^n n^{-\alpha_P+o(1)}, \tag{23.1}$$

$$s_n = \Gamma_S^n n^{-\alpha_S+o(1)}, \tag{23.2}$$

$$b_n = \Gamma_S^n n^{-\alpha_S+o(1)}. \tag{23.3}$$

All three sequences are eventually positive at every integer index. Their ordinary generating functions are not D -finite over $\mathbb{C}(z)$. For each triple (a_n, Γ, α) equal to $(p_n, \Gamma_P, \alpha_P)$, $(s_n, \Gamma_S, \alpha_S)$ or $(b_n, \Gamma_S, \alpha_S)$, and the first threshold $N_a(y) = \min\{n \geq 0 : a_n \geq y\}$,

$$N_a(y) = \frac{\log y}{\log \Gamma} + \frac{\alpha}{\log \Gamma} \log \log y + o(\log \log y) \quad (y \rightarrow \infty). \quad (23.4)$$

No monotonicity assumption is needed. The b_n conclusions use the six-face injection in Section 29.

[write, 2026-10-05] Every conclusion of this theorem is proved in a stronger form in Parts I–II: (23.1) and (23.2) are (1.3), consequences of the Θ bounds of Theorem 1.1 and of the equivalents of Theorem 11.1; (23.3) follows from the equivalent (11.4); non- D -finiteness of the three generating functions is Corollary 9.4; and Lemma 9.5 gives (23.4) with error $O(1)$, Theorem 20.1 with the next constant. Theorem 23.1 is therefore a second route, by a different method (Table 2).

The proof separates into a general probabilistic result (Section 24), exact model checks (Sections 25–26), and arithmetic and inverse arguments (Sections 27–28). Section 29 proves the explicit six-face injection and resulting all-index theorem for A377921.

What was already known. Theorem 24 of [1] proves the exponential growth constants $9/2$ and $16/3$. Its Conjecture 25 proposes stronger full coefficient equivalents with the powers appearing above. Equations (23.1)–(23.2) establish the exponents on the logarithmic scale only. They leave the positive multiplicative constants and the full equivalents unresolved. The literature review in Section 30 is scoped; it is not a certificate of priority or an exhaustive novelty claim.

Numerical orientation. The exponents are approximately 4.23 and 6.08, respectively. All proofs and all exact checks use the symbolic expressions, not these decimals. The stronger inverse precision $O(1)$, further inverse terms and coefficient transseries are not consequences of this report.

[write, 2026-10-05] Both paragraphs describe Report 112 alone. The positive constants and the full equivalents left unresolved here are proved in Part II (Theorem 11.1); the inverse with error $O(1)$ is Lemma 9.5, and its next additive constant is in Theorem 20.1. No coefficient transseries is proved anywhere in this report.

24 A cone theorem with finite phases and exponential tails

24.1 Hypotheses and statement

Consider a translation-invariant Markov-additive chain (S_n, C_n) on $\mathbb{Z}^2 \times E$, with E finite and

$$p_{cd}(v) = \mathbb{P}(S_{n+1} - S_n = v, C_{n+1} = d \mid C_n = c), \quad \sum_{d,v} p_{cd}(v) = 1.$$

Let $P_{cd} = \sum_v p_{cd}(v)$. The spatial coordinates need not have independent or bounded increments. Assume the following.

- (H1) P is primitive. Its stationary distribution π has all entries strictly positive.
- (H2) For some $\lambda > 0$, $\sup_c \mathbb{E}_c e^{\lambda |S_1 - S_0|} < \infty$. Any fixed norm may be used.
- (H3) The stationary drift is zero and the effective covariance Σ is positive definite.

- (H4) For the Fourier matrix $F(t)_{cd} = \sum_v p_{cd}(v)e^{it \cdot v}$, $\text{rad } F(t) < 1$ at every nonzero point of $\mathbb{R}^2/(2\pi\mathbb{Z})^2$. At zero the only unit eigenvalue is the simple Perron eigenvalue 1.
- (H5) Fix endpoint phases a, b at spatial origin. Put $W = \Sigma^{-1/2}$ and let $\mathcal{K} = W\mathbb{R}_+^2$ be the whitened wedge, with angle $\theta \in (0, \pi)$. There is a compact angular interval of positive length $J \subset (0, \theta)$ such that, for every $R > 0$, a finite positive-probability legal path from $(0, a)$ reaches a point x with $|Wx| \geq R$ and $\arg(Wx) \in J$. The same holds from $(0, b)$ for the stationary dual, defined by

$$p_{dc}^*(-v) = \frac{\pi_c}{\pi_d} p_{cd}(v). \quad (24.1)$$

Legal means that every visited lattice point lies in the *closed* quadrant. Seed paths in (H5) have no prescribed length, no probability uniform in R , and no requirement to stay inside their terminal radius. For each tolerance in the proof a single sufficiently deep seed is fixed before n tends to infinity.

Let $K_n((x, c), (y, d))$ denote the killed transition kernel, and let $\tau = \inf\{n \geq 0 : S_n \notin \mathbb{R}_+^2\}$. Write

$$\rho = \frac{\Sigma_{12}}{\sqrt{\Sigma_{11}\Sigma_{22}}}, \quad \theta = \arccos(-\rho), \quad \nu = \frac{\pi}{\theta}.$$

The angle formula follows by computing the angle between We_1 and We_2 from $W^T W = \Sigma^{-1}$.

Theorem 24.1 (Finite-phase logarithmic cone excursions). *Under (H1)–(H5), at the specified phases,*

$$K_n((0, a), (0, b)) = n^{-1-\nu+o(1)}. \quad (24.2)$$

Under (H1)–(H4) alone, for each fixed starting phase and every $\epsilon > 0$,

$$\mathbb{P}_{(0,c)}(\tau > n) \leq C_\epsilon n^{-\nu/2+\epsilon}. \quad (24.3)$$

The same upper bound holds for the dual chain.

The seed condition is substantive. An unrestricted centered chain with nondegenerate covariance can have a boundary phase from which no legal interior excursion is possible. The theorem makes no claim for such an endpoint. It also does not assert a constant equivalent in (24.2).

[write, 2026-10-05] Part I (Section 1) says that it does not posit a general cone theorem for Markov-modulated walks and that its argument uses a geometric property of the two models, the one-unit cycle buffer of Lemma 4.1. Theorem 24.1 is a general theorem of that kind at the logarithmic scale: it needs no regeneration, no cycle buffer and no iid cone input, and it allows unbounded steps. Its price is the factor $n^{o(1)}$. For the two models of this report Parts I–II give Θ bounds and equivalents; whether (H1)–(H5) alone give either is open (Question Q1). Section 22.5 names two other chains of the same kind.

24.2 Corrector and unrestricted limits

Let $m(c) = \mathbb{E}_c(S_1 - S_0)$. Stationary centering solves the finite Poisson equation

$$(I - P)h = m, \quad \sum_c \pi_c h(c) = 0.$$

The process

$$Z_n = S_n + h(C_n) - h(C_0)$$

is a martingale. Its increments have a uniform exponential moment. If $V(c)$ is their conditional covariance, then $\Sigma = \sum_c \pi_c V(c)$. The finite-state ergodic theorem gives the rescaled predictable quadratic variation $t\Sigma$. Its jumps are uniformly bounded before the n^{-1} scaling, while

$$\mathbb{E} \max_{j \leq n} |\Delta Z_j|^2 = O(\log^2(n+2)).$$

The last estimate follows by integrating the union bound for exponential tails. Thus the maximum rescaled squared jump tends to zero and the martingale functional central limit theorem applies; see [3, Theorem 2.1(ii)]. Since h is bounded,

$$R^{-1}(S_{\lfloor R^2 t \rfloor} - S_0) \implies B_\Sigma(t). \quad (24.4)$$

The convergence is uniform over the finite phase set. Translation invariance allows varying initial positions whose rescaled positions converge. Polygonal and step interpolation differ by $o_{\mathbb{P}}(1)$, since $R^2 \mathbb{P}(|\Delta S| > \epsilon R) \rightarrow 0$.

The exponential moment makes F analytic near the real Fourier torus. Its simple eigenvalue near zero and spectral projection satisfy

$$\lambda(t) = 1 - \tfrac{1}{2} t^T \Sigma t + O(|t|^3), \quad \Pi(t) \longrightarrow \mathbf{1}\pi.$$

For example, the Hessian of the log Perron eigenvalue agrees with the martingale covariance because the bounded corrector does not change asymptotic variance. Away from zero, (H4), compactness and finite-matrix spectral estimates bound powers exponentially. Near zero, the positive definite quadratic term gives a Gaussian bound. Fourier inversion, rescaling $t = w/\sqrt{n}$, and dominated convergence therefore give

$$\sup_{z, c, d} |n \mathbb{P}_c(S_n - S_0 = z, C_n = d) - \pi_d g_\Sigma(z/\sqrt{n})| \longrightarrow 0, \quad (24.5)$$

where g_Σ is the centered normal density with covariance Σ . Uniformity in z follows because the Fourier factors have modulus one. In particular the unrestricted kernel is at most C/n everywhere and at least c/n on compact sets of scaled displacements.

The dual preserves centering, exponential tails and effective covariance. Its Fourier matrix is $\text{diag}(\pi)^{-1} F(-t)^T \text{diag}(\pi)$, so it preserves (H4). Reversing an individual path gives

$$\pi_c K_j((x, c), (y, d)) = \pi_d K_j^*((y, d), (x, c)). \quad (24.6)$$

24.3 Exact-time bridges in the interior

Lemma 24.2. *For every compact $A \subset (0, \infty)^2$ and $0 < u < v < \infty$, there exists $c_A > 0$ such that, for all sufficiently large n ,*

$$K_m((x, c), (y, d)) \geq c_A/n \quad (24.7)$$

uniformly for $un \leq m \leq vn$, $x/\sqrt{n}, y/\sqrt{n} \in A$ and all phases.

Proof. Condition the unrestricted chain on its exact endpoint. Through time $k = \lfloor tm \rfloor$, for fixed $t < 1$, its density relative to the unconditioned chain is the remaining endpoint kernel divided by the full endpoint kernel. Equation (24.5) bounds this density uniformly and makes it converge on compact scaled positions to the Gaussian ratio defining a Brownian bridge. The FCLT, tightness and compact truncation give the bridge limit on $[0, t]$. Reversal gives the analogous terminal portion through a dual bridge. Tightness on $[0, 2/3]$ and $[1/3, 1]$ yields tightness on the whole interval, as every pair of sufficiently close times lies in one of these intervals.

The segment between two points of A stays a uniformly positive distance from the sides of the quadrant. A Brownian bridge has a uniformly positive probability of staying in a sufficiently thin open tube around this segment, uniformly over the compact endpoint and time parameters. Portmanteau, followed by the unrestricted endpoint lower bound c/n , proves (24.7). Convexity makes polygonal containment equivalent to containment of all visited vertices, even when a jump is long. The bridge tightness argument already controls rare large jumps under conditioning; no extra conditional tail assumption is used. $\square$

24.4 Brownian crossing bounds and relative overshoot

Use whitened coordinates $X_n = WS_n$. Both chains have a common bound

$$\mathbb{P}_c(|\Delta X| > r) \leq Ae^{-\kappa r}. \quad (24.8)$$

The FCLT also supplies constants $A_0, b_0, R_1 > 0$ such that, from every point in the radius- R disk and every phase,

$$\mathbb{P}_x \left(\max_{j \leq t} |X_j| \leq R \right) \leq A_0 e^{-b_0 t / R^2} \quad (R \geq R_1). \quad (24.9)$$

Indeed an unrestricted displacement larger than $3R$ in $\lceil R^2 \rceil$ steps forces an exit, with probability bounded below by the FCLT uniformly in phase. Iterate the Markov property.

For planar Brownian motion in the wedge, the probability of reaching radius q before a side, starting at radius one, is at most $C_B q^{-\nu}$ uniformly in the starting angle. If the starting angle lies in a fixed compact interval J inside the wedge, the probability of exiting through the interior of the outer arc with angle in J is at least $c_B q^{-\nu}$ for all sufficiently large q . The constants are independent of q . One direct verification is the sine expansion of the Dirichlet solution with outer boundary data f :

$$H_f(r, \phi) = \sum_{k \geq 1} b_k (r/q)^{k\nu} \sin(k\nu\phi), \quad b_k = \frac{2}{\theta} \int_0^\theta f(\psi) \sin(k\nu\psi) d\psi. \quad (24.10)$$

For $f = 1$, the sum is $O(q^{-\nu})$ at $r = 1$. For $f = \mathbf{1}_J$, the first coefficient is positive and $\min_{\phi \in J} \sin(\nu\phi) > 0$, whereas the remaining sum is $O(q^{-2\nu})$. Brownian disk confinement truncates the successful exit time at $T(q)q^2$ while retaining a fixed fraction of its mass.

Let $\sigma_r = \inf\{j : |X_j| \geq r\}$. There are constants $c, C > 0$ independent of sufficiently large fixed q , then choices $T = T(q)$ and $R_0 = R_0(q)$, such that every start with $R \leq |X_0| \leq R + \sqrt{R}$, $R \geq R_0$, satisfies

$$\mathbb{P}_x(\sigma_{qR} < \tau) \leq Cq^{-\nu}. \quad (24.11)$$

If additionally its angle lies in J , then

$$\mathbb{P}_x \left(\begin{array}{l} \sigma_{qR} < \tau, \quad \sigma_{qR} \leq T(qR)^2, \\ \arg X_{\sigma_{qR}} \in J, \quad \max_{1 \leq j \leq \sigma_{qR}} |\Delta X_j| \leq \sqrt{R} \end{array} \right) \geq cq^{-\nu}. \quad (24.12)$$

Here the maximum concerns increments *after this stage begins*.

For completeness, the uniform transfer from Brownian motion is as follows. The permitted starting radii divided by R tend uniformly to one. On each fixed scaled horizon the event of reaching the outer circle while staying in the closed wedge until that time is closed. At a limiting side start its Brownian probability is zero, since Brownian motion immediately enters the exterior half-plane. At an interior start the harmonic bound applies. Compact subsequences of starting points and the finite phase set make the upper estimate uniform. Hits after the chosen horizon are bounded by (24.9). Choose $T(q)$ first and then $R_0(q)$ so each additional error is at most a fixed multiple of $q^{-\nu}$.

For the lower estimate, the exit event through the interior of J is an almost-sure continuity event: Brownian motion hits no corner or arc endpoint and immediately crosses a smooth exit circle. On a successful path it stays a positive distance from the sides up to and just after exit. A strict time cutoff with no atom may be chosen. The FCLT gives the lower estimate before imposing the jump restriction. The extra excluded probability is at most

$$\lceil Tq^2R^2 \rceil Ae^{-\kappa\sqrt{R}} = o_R(1). \quad (24.13)$$

For the already fixed q , it is smaller than the allotted fraction of $c_B q^{-\nu}$ for all large R . The terminal radius then lies in $[qR, qR + \sqrt{R}] \subset [qR, qR + \sqrt{qR}]$, precisely the allowance required for the next stage. The probabilities above are always under the original chain; its transition law is never conditioned or truncated.

24.5 Upper survival from good past events

Fix $\eta \in (0, \nu)$ and choose one sufficiently large $q \geq 2$ so that

$$Cq^{-\nu} \leq q^{-\nu+\eta}, \quad cq^{-\nu} \geq q^{-\nu-\eta}. \quad (24.14)$$

After $T(q), R_0(q)$ are fixed, enlarge R_0 so $\sqrt{R} < (q-1)R$ for $R \geq R_0$. Take $M > \nu + 2$ and D sufficiently large. Put

$$L_n = D \log(n+2), \quad \mathcal{G}_n = \{|\Delta X_k| \leq L_n \text{ for } 1 \leq k \leq n\}.$$

By (24.8),

$$\mathbb{P}(\mathcal{G}_n^c) \leq nAe^{-\kappa L_n} = O(n^{-M}). \quad (24.15)$$

For this horizon define $r_0(n) = \max\{R_0, L_n^2\}$ and $r_j = q^j r_0(n)$, and let σ_j be the first global hit of radius r_j . The relevant measurable *past* event is

$$A_j = \{\sigma_j \leq n, \sigma_j < \tau, |\Delta X_k| \leq L_n \text{ for all } k \leq \sigma_j\}.$$

On A_j the overshoot is at most $L_n \leq \sqrt{r_j}$. A good jump cannot skip the next radius, so $A_{j+1} \subset A_j$. At σ_j , drop all future jump restrictions and the time deadline. The probability of the required next crossing before killing is at most $q^{-\nu+\eta}$ by (24.11). Strong Markov gives

$$\mathbb{P}(A_{j+1}) \leq q^{-\nu+\eta} \mathbb{P}(A_j), \quad \mathbb{P}(A_J) \leq q^{-(\nu-\eta)J}. \quad (24.16)$$

This step does not condition the process on the future event $\mathcal{G}_n$. All future crossing estimates use its original law.

Take $R_n = \sqrt{n/(B \log(n+2))}$ with fixed B so large that the confinement bound is $O(n^{-M})$, and choose J with $r_J \leq R_n < r_{J+1}$. A surviving path either has a bad jump, remains inside radius R_n , or belongs to A_J . Therefore

$$\begin{aligned} \mathbb{P}(\tau > n) &\leq O(n^{-M}) + C(R_n/r_0(n))^{-\nu+\eta} + A_0 e^{-b_0 n/R_n^2} \\ &\leq n^{-(\nu-\eta)/2} (\log n)^{O(1)} + O(n^{-M}). \end{aligned} \quad (24.17)$$

Choosing η sufficiently small proves (24.3) for any prescribed $\epsilon > 0$. The same argument applies to the dual. No eventual escape estimate from a fixed small disk is needed.

24.6 Lower escapes and exact-time concatenation

After η, q, T, R_0 have been fixed, choose one legal seed for each chain whose terminal radius r_0 is at least R_0 and whose terminal angle lies in J . The two seed radii may differ. Each seed length, probability and largest visited radius is a fixed finite constant. For each chain separately use radii $r_j = q^j r_0$ and successively accept the events in (24.12). Their total mass through radius r_J is at least the seed probability times $q^{-(\nu+\eta)J}$. The total elapsed time is at most

$$t_{\text{seed}} + T \sum_{j=1}^J r_j^2 \leq t_{\text{seed}} + \frac{Tr_J^2}{1 - q^{-2}}. \quad (24.18)$$

Choose fixed $\delta > 0$ with $T\delta^2/(1 - q^{-2}) \leq 1/8$, and for each seed use the largest J with $r_J \leq \delta\sqrt{n}$. Then $r_J > \delta\sqrt{n}/q$, the total time is at most $n/4$ for large n , and endpoints divided by $\sqrt{n}$ lie in one fixed compact subset of the open quadrant. The forward and dual masses F_n, G_n obey

$$F_n, G_n \geq c_\eta n^{-(\nu+\eta)/2}. \quad (24.19)$$

For large n , the chosen final radius exceeds the largest radius visited anywhere in the corresponding fixed seed. Hence an accepted prefix ends at the *first global hit* of its final radius, even if the seed had previously overshoot its own terminal radius. This is why no monotone-radius condition was imposed on the seeds.

Join any accepted forward prefix of length $t \leq n/4$ to the reversal of any accepted dual prefix of length $u \leq n/4$ with a killed bridge of exact length $n - t - u$. Lemma 24.2 bounds its mass below by c/n . Reversal (24.6) changes the suffix weight by π_b/π_d , bounded above and below. There is no overcounting: the first forward hit of the prescribed forward radius and first backward hit of the prescribed backward radius recover the two prefixes uniquely. Their lengths cannot overlap. Consequently

$$K_n((0, a), (0, b)) \geq (c/n)F_n G_n \geq c_\eta n^{-1-\nu-\eta}. \quad (24.20)$$

For the upper bound, split n into three comparable integer pieces. Bound the middle unrestricted kernel by C/n using (24.5), and bound the outer pieces by forward and dual survival using (24.6). Equation (24.3) yields

$$K_n((0, a), (0, b)) \leq C_\epsilon n^{-1-\nu+\epsilon}. \quad (24.21)$$

Together these bounds prove Theorem 24.1. They retain every integer time and every intervening jump. In particular, the result is not a random-time or subsequence estimate.

25 Polyhedral orientations and A377922

25.1 Original walk and quotient coordinates

For $n \geq 1$, the exact walk correspondence in [1, Proposition 13] counts p_n by length- n walks from $(0, 0)$ to the horizontal axis, remaining in $\mathbb{Z}_{\geq 0}^2$. An SE step is $(1, -1)$. A face step is $(-i, j)$ with $i, j \geq 0$ and $i + j$ even; when the current height is even it must have $j \geq 1$, and when the current height is odd it must have $i \geq 1$. These conditions keep $x \equiv y \pmod{2}$.

Encode this physical parity by

$$x = 2U + c, \quad y = 2V + c, \quad c \in \{0, 1\}. \quad (25.1)$$

This is a genuine two-phase chain on a full quotient lattice. The physical inequalities $x, y \geq 0$ are exactly $U, V \geq 0$ for either phase, including the boundary.

Set $z = 1/3$, $\Gamma_P = 9/2$ and $D = (1 - z/u)(1 - zv)$. The normalized transition generating matrix is

$$P(u, v) = \Gamma_P^{-1} \begin{pmatrix} zv/D & z^{-1}v^{-1} + z/(uD) \\ u/z + zv/D & z/(uD) \end{pmatrix}. \quad (25.2)$$

A coefficient $u^r v^s$ records quotient displacement (r, s) . The tilted physical weight of an elementary step is $z^{(\Delta y - \Delta x)/2} / \Gamma_P$. The factor $1/2$ in this exponent is required by the two-coordinate quotient. In particular it must be retained in telescoping path weights.

An elementary probabilistic reconstruction makes positivity and every phase explicit. Choose SE with probability $2/3$ or a face with probability $1/3$. For a face, choose destination phase d uniformly, and independently choose $I, J \geq 0$ with $\mathbb{P}(I = i) = \mathbb{P}(J = i) = (2/3)(1/3)^i$. Then

$$\begin{aligned} \text{SE: } d &= 1 - c, \quad \Delta = (c, c - 1), \\ \text{face: } \Delta &= (-I - d, J + 1 - d). \end{aligned}$$

This reproduces (25.2) directly. Thus

$$P(1, 1) = \begin{pmatrix} 1/6 & 5/6 \\ 5/6 & 1/6 \end{pmatrix}, \quad \pi = (1/2, 1/2).$$

25.2 Centering and covariance

The conditional drifts and mean-zero corrector are

$$m_0 = (-1/3, -1/3) = -m_1, \quad h_0 = (-1/5, -1/5) = -h_1.$$

They satisfy $(I - P(1, 1))h = m$. The covariance of the corrected increment $T = \Delta + h_d - h_c$ under stationarity is

$$\Sigma_P = \begin{pmatrix} 4/5 & -9/20 \\ -9/20 & 4/5 \end{pmatrix}, \quad \rho_P = -9/16. \quad (25.3)$$

Its eigenvalues are $7/20$ and $5/4$.

Here is a direct second-moment calculation independent of Perron root differentiation. On a face, $T = (-I - B, J + 1 - B)$ with $B = (3d + 2c)/5$. Conditional on a face in stationarity, c, d are independent uniform bits, $\mathbb{E}B = 1/2$ and $\text{Var } B = 13/100$. Hence the face mean is $(-1, 1)$, each diagonal covariance is $22/25$, and the off-diagonal covariance is $13/100$. On SE, $T = (2/5 + c/5, -3/5 + c/5)$, its mean is $(1/2, -1/2)$, and every covariance entry is $1/100$. Mixing the *second moments* with probabilities $1/3$ and $2/3$ gives (25.3); the total mean is zero.

The executable certificate also computes the Hessian at $(1, 1)$ of $\log \lambda_P(u, v)$ with logarithmic derivatives $u\partial_u, v\partial_v$, where λ_P is the Perron root. This is a separate exact verification of the same effective covariance.

The physical asymptotic covariance is $4\Sigma_P$, because the phase coordinate offsets are bounded. The matrix displayed in the source's asymptotic calculation is Γ_P times this physical stochastic covariance. Differentiating an unnormalized Perron root rather than its normalized logarithm produces that scale. Correlation, and thus the cone angle, are unchanged; the scales must not be interchanged in a stochastic normalization.

[write, 2026-10-05] Part I makes the same observation after (3.7), in physical units: its Σ_P is the $4\Sigma_P$ of this paragraph, and [1, Section 4.2] displays $\mu_P \Sigma_P$. The kernel (25.2) is K_P of (2.4) in half-scale variables (both give the face with $c = d = 0$, $I = i$, $J = j$ the weight $(2/27)3^{-i-j}$); the drifts, correctors and covariances correspond as in Remark 22.1.

25.3 Exponential tails and exact aperiodicity

The quotient increments are affine functions of geometric variables and bounded phases. Thus $\sup_c \mathbb{E}_c e^{t\|\Delta\|_1} < \infty$ for $0 < t < \log 3$. The bounded corrector and stationary dual preserve a positive exponential-moment neighborhood.

To check the full lattice and time condition (H4), suppose the Fourier matrix has a unit-modulus eigenvalue λ at frequency t . Irreducibility and equality in the triangle inequality give unit phases a_c with

$$e^{it \cdot \delta} a_d = \lambda a_c \quad \text{on every supported edge } (c, \delta, d). \quad (25.4)$$

The supported phase-0 loops $(0, 1), (0, 2)$ force $e^{it_2} = e^{2it_2} = \lambda$, hence $t_2 = 0$ modulo 2π and $\lambda = 1$. The phase-1 loop $(-1, 0)$ forces $t_1 = 0$. This proves (H4) on the full quotient torus. An assertion of full lattice aperiodicity in the redundant physical coordinates would be incorrect; their parity is precisely what the phase representation removes.

25.4 Legal seeds and endpoint normalization

For any integer $H \geq 1$, a forward seed from quotient $(0, 0, 0)$ is: take a phase-0 loop $(0, 2H)$, then repeat $0 \rightarrow 1 : (0, -1)$ followed by $1 \rightarrow 0 : (1, 0)$, H times. It reaches $(H, H, 0)$ legally.

The dual from $(0, 0, 1)$ takes $2H$ phase-1 east loops $(1, 0)$, then repeats $1 \rightarrow 0 : (0, 1)$ followed by $0 \rightarrow 1 : (-1, 0)$, H times. It reaches $(H, H, 1)$ legally. Every listed edge has positive probability. After whitening, the ray through (H, H) has a fixed strictly interior angle. This proves (H5), with an interval of positive length surrounding that angle. The seeds may visit a greater radius before their terminal point, which is permitted.

Let e_L count legal original paths of length L from $(0, 0)$ to $(1, 1)$. The endpoint tilt equals one, and therefore

$$e_L = \Gamma_P^L K_L((0, 0, 0), (0, 0, 1)). \quad (25.5)$$

The ordered triples here mean (U, V, c) . For $n \geq 1$,

$$e_{n-1} \leq p_n \leq e_{n+1}. \quad (25.6)$$

For the lower injection append SE to a path ending at $(1, 1)$; its endpoint becomes $(2, 0)$. For the upper injection, a nonempty legal path ending on the horizontal axis cannot end at $(0, 0)$: an incoming SE would require $x = -1$, and an incoming face at even height zero is forbidden. Its endpoint is $(2i + 2, 0)$ for some $i \geq 0$. Append the allowed face $(-2i - 1, 1)$ to reach $(1, 1)$. Removing that last step recovers the original path, proving injectivity.

Theorem 24.1, (25.5) and the two fixed shifts in (25.6) prove (23.1). This argument avoids summing boundary probabilities with unequal endpoint tilt weights.

[write, 2026-10-05] The sandwich (25.6) is new to this report. Part I uses the tilted endpoint sum (2.6), which needs the uniform endpoint bound (8.6); Part II sums the same series with amplitudes (19.1). Combined with Theorem 18.1 at the endpoints $(0, 0)$ and $(1, 1)$, the sandwich gives only Θ for p_n , because the two fixed shifts change μ_P^n by the factors $\mu_P^{\pm 1}$.

26 Schnyder labelings and A377920

26.1 The original weighted walk and its time parameter

The source's auxiliary subfamily s'_n imposes nonisolation at the two outer vertices labeled G . Its walk correspondence [1, Proposition 14], for genuine sizes $n \geq 1$, uses paths from physical $(0, 2)$ to

$(2, 0)$ in the quadrant with exactly $n + 2$ SE steps. Face multiplicities are important. With $\ell, r \geq 0$, a face $(-i, j)$ has weight $\binom{\ell+r}{r}$ for

$$(i, j) = (2\ell + 2, 2r + 2),$$

or for

$$(i, j) = \begin{cases} (2\ell + 1, 2r + 3), & \text{current height even,} \\ (2\ell + 3, 2r + 1), & \text{current height odd.} \end{cases}$$

Every face strictly lowers x and strictly raises y .

One aggregate is one SE step followed by all succeeding faces up to the next SE. Its duration as a sequence of elementary steps is variable, but it contains exactly one SE. This aggregate count is the required deterministic size parameter. Treating a face micro-decomposition or its random duration as the counting time would change the enumerative problem.

For physical generating variables a, b put

$$F(a, b) = \frac{1}{1 - a^{-2} - b^2} \begin{pmatrix} a^{-2}b^2 & a^{-1}b^3 \\ a^{-3}b & a^{-2}b^2 \end{pmatrix}, \quad J(a, b) = \begin{pmatrix} 0 & a/b \\ a/b & 0 \end{pmatrix}. \quad (26.1)$$

Then the aggregate matrix is

$$A(a, b) = J(a, b)(I - F(a, b))^{-1}. \quad (26.2)$$

The matrix order is essential: SE changes parity before the following faces. The nonnegative expansion $J \sum_{k \geq 0} F^k$ retains all face multiplicities. At $a = 2, b = 1/2$, every entry of F is $1/8$, its row sums are $1/4$, and the row sums of A are $\Gamma_S = 16/3$.

26.2 Shifted quadrant and exact killing equivalence

An original path must start with SE, since no face is legal at $x = 0$, and end with SE, since faces strictly raise y . Deleting its last SE leaves $n + 1$ aggregates from $(0, 2)$ to $(1, 1)$. Aggregate endpoints lie in the *shifted* physical quadrant $x \geq 0, y \geq 1$. Introduce

$$x = 2U + c, \quad y = 2V + 2 - c, \quad c \in \{0, 1\}. \quad (26.3)$$

For both phases the shifted condition is exactly $U, V \geq 0$. The two endpoints become quotient $(0, 0, 0)$ and $(0, 0, 1)$.

Within an aggregate, SE raises x and lowers y by one, after which faces only lower x and raise y . Its minimum x is at an endpoint; its minimum y is immediately after SE. Shifted-quadrant endpoints therefore guarantee that every elementary point stays in the original quadrant. Conversely, every nonfinal aggregate endpoint precedes a legal SE and hence has $y \geq 1$; the prescribed final endpoint has $y = 1$. This proves the equivalence in both directions, including every intermediate face point. There is no hidden within-block boundary crossing after aggregation.

26.3 The rational two-phase kernel

The normalized quotient kernel is

$$G_{cd}(u, v) = \Gamma_S^{-1} A_{cd}(2\sqrt{u}, \sqrt{v}/2) u^{(c-d)/2} v^{(d-c)/2}. \quad (26.4)$$

The phase factors remove the apparent half-integral exponents. A simpler exact rational form is obtained by setting

$$C(u, v) = \frac{3}{8(8u - 2uv - v - 2)} = \frac{3/(64u)}{1 - v/4 - 1/(4u) - v/(8u)}.$$

Then

$$G(u, v) = \begin{pmatrix} C & 3/4 + vC \\ 3u/(4v) + uC & uvC \end{pmatrix}. \quad (26.5)$$

The displayed geometric expansion proves nonnegative Laurent coefficients directly. In particular

$$G(1, 1) = \begin{pmatrix} 1/8 & 7/8 \\ 7/8 & 1/8 \end{pmatrix}, \quad \pi = (1/2, 1/2).$$

The conditional means and corrector are

$$m_0 = (-1/2, 3/8) = -m_1, \quad h_0 = (-2/7, 3/14) = -h_1.$$

The resulting effective covariance is

$$\Sigma_S = \begin{pmatrix} 9/7 & -22/21 \\ -22/21 & 9/7 \end{pmatrix}, \quad \rho_S = -22/27. \quad (26.6)$$

Its eigenvalues are 5/21 and 7/3. The physical covariance is $4\Sigma_S$; the source's displayed matrix is Γ_S times this normalized stochastic matrix. Again the correlation is identical.

For an independent moment derivation, let N be the number of faces in one tilted aggregate: $\mathbb{P}(N = k) = (3/4)(1/4)^k$. Each face contributes (ℓ, r) with law

$$\mathbb{P}(\ell, r) = \frac{1}{2} \binom{\ell + r}{r} 4^{-\ell - r}.$$

Its marginal means are 1/2, marginal variances 3/4, and covariance 1/4. Let L, R be the sums of these two coordinates over the N faces. If $N = 0$, then $d = 1 - c$; otherwise the terminal phase d is a uniform bit independent of c, N, L, R . Telescoping gives

$$\Delta = (c - N - L, N + R + d - 1).$$

With $X = N + L$, $Y = N + R$, $B = (3c + 4d)/7$, the corrected increment is $T = (-X + B, Y + B - 1)$. Since $\mathbb{E}N = 1/3$ and $\text{Var } N = 4/9$,

$$\mathbb{E}X = \mathbb{E}Y = 1/2, \quad \text{Var } X = \text{Var } Y = 5/4, \quad \text{Cov}(X, Y) = 13/12.$$

Also $\mathbb{E}(B \mid N) = 1/2$, $\text{Var } B = 1/28$ and $\text{Cov}(B, X) = \text{Cov}(B, Y) = 0$. These identities give (26.6). The exact checker independently compares the result with the log Perron-root Hessian and the Poisson-corrector calculation.

[**write, 2026-10-05**] G is K_S of (2.11) in half-scale variables, and (26.6) is (3.7) divided by 4 (Remark 22.1). The aggregate time, the shifted quadrant and its equivalence with physical survival are Part I's (Lemma 2.1), proved here a second time.

26.4 A moment bound that retains the whole aggregate

Let L_{tot} be the sum of physical ℓ^1 lengths of every elementary step of one aggregate. At the critical tilt, the sum of weighted face contributions in each row, with extra factor $e^{t \text{length}}$, is

$$f(t) = \frac{e^{4t}}{8(1 - e^{2t}/2)}.$$

SE contributes $(3/4)e^{2t}$ after normalization. Summing the number of faces gives the exact identity

$$\mathbb{E}_c e^{tL_{\text{tot}}} = \frac{(3/4)e^{2t}}{1 - f(t)}, \quad f(t) < 1, \quad e^{2t} < 2. \quad (26.7)$$

For $t = \log(6/5)$, $e^{2t} = 36/25$ and $f(t) = 162/175 < 1$; indeed the displayed moment equals $189/13$. The aggregate displacement norm is bounded by L_{tot} , and quotient coordinates differ by division by two and a bounded phase shift. Thus (H2) holds uniformly, including all within-aggregate lengths rather than only a formal rational-kernel derivative.

[write, 2026-10-05] This explicit certificate is new to the report. Part I obtains the exponential moment of a return cycle from analyticity of (3.9) at $(1, 1, 1)$. The arithmetic was rechecked at the write: $e^{2t} = 36/25$, $f(t) = (1296/625)/(8 \cdot 7/25) = 162/175$, and $(3/4)(36/25)/(13/175) = 189/13$.

26.5 Aperiodicity and legal seeds

Supported diagonal increments include phase-0 loops $(-1, 0)$, $(-2, 0)$ and a phase-1 loop $(0, 1)$. Equation (25.4) first forces $t_1 = 0, \lambda = 1$, then $t_2 = 0$ modulo 2π . Hence (H4) holds in exact aggregate time.

A forward seed from $(0, 0, 0)$ takes $0 \rightarrow 1 : (0, 0)$, then $2H$ phase-1 north loops, followed by H repetitions of $1 \rightarrow 0 : (1, -1)$ and $0 \rightarrow 1 : (0, 0)$. It ends at $(H, H, 1)$. A dual seed from $(0, 0, 1)$ takes $1 \rightarrow 0 : (0, 0)$, then $2H$ phase-0 east loops, followed by H repetitions of $0 \rightarrow 1 : (-1, 1)$ and $1 \rightarrow 0 : (0, 0)$. It ends at $(H, H, 0)$. All edges have positive probability and all coordinates stay nonnegative. The resulting fixed interior ray proves (H5).

26.6 Endpoint factor and transfer to all labelings

The truncated path has physical displacement $(1, -1)$, and its telescoping tilt is $2^1(1/2)^{-1} = 4$. For genuine sizes $n \geq 1$,

$$K_{n+1}((0, 0, 0), (0, 0, 1)) = 4\Gamma_S^{-n-1}s'_n. \quad (26.8)$$

The positive fixed-shift relation from [1, Proposition 19] is

$$s_n = s'_n + 2s'_{n-1} + s'_{n-2} \quad (n \geq 4). \quad (26.9)$$

Applying Theorem 24.1 to (26.8) and then using (26.9) proves (23.2). Each positive term has the same logarithmic power; no cancellation or random-time replacement is involved.

[write, 2026-10-05] (26.8) is (2.13), and (26.9) is (2.8); see Section 22.3 for the two citations of [1].

In finite enumeration a formal auxiliary walk value $s'_0 = 1$ is convenient. It is an initialization outside the genuine size range of the source bijection, not a count of a size-zero nonempty dissection. The actual initial values are $s_0, s_1, s_2, s_3 = 0, 0, 3, 2$; (26.9) is applied only from $n = 4$.

27 Non-D-finiteness from logarithmic powers

27.1 Irrationality and the needed derivative order

If θ/π were rational, $e^{i\theta}$ would be a root of unity and $2\cos\theta$ would be an algebraic integer. In our two cases the latter numbers are $9/8$ and $44/27$, rational nonintegers. Thus both θ/π , and hence both $\alpha = 1 + \pi/\theta$, are irrational. Elementary angle comparisons give

$$4 < \alpha_P < 5, \quad 6 < \alpha_S < 7.$$

For the second interval, $\pi/6 < \arccos(22/27) < \pi/5$: the nontrivial upper-angle inequality uses $22/27 > (1 + \sqrt{5})/4$, equivalent to $61/27 > \sqrt{5}$, and $3721 > 3645$ verifies its square. The first interval follows from $\pi/4 < \arccos(9/16) < \pi/3$.

27.2 A positive derivative detects the real singular exponent

Let a_n denote either sequence and $F(z) = \sum_{n \geq 0} a_n z^n$. Put $R = \Gamma^{-1}$ and $H(t) = F(Rt)$. Take $m = \lfloor \alpha \rfloor$, namely $m = 4$ or 6 . The coefficients of $H^{(m)}$ are positive for large indices and have size $n^{m-\alpha+o(1)}$, with $m - \alpha \in (-1, 0)$. For any real $s > -1$,

$$\sum_{n \geq 1} n^s t^n \asymp (1-t)^{-s-1} \quad (t \uparrow 1). \quad (27.1)$$

To see this, write $t = e^{-u}$. Restricting to $n \in [1/u, 2/u]$ gives the lower bound; partitioning into intervals $[k/u, (k+1)/u]$ gives an upper bound from an exponentially convergent sum and the integrability of x^s near zero. Finally $u \sim 1-t$.

For every sufficiently small $\epsilon > 0$, positive termwise comparison with exponents $m - \alpha \pm \epsilon$ yields

$$c_\epsilon (1-t)^{\alpha-m-1+\epsilon} \leq H^{(m)}(t) \leq C_\epsilon (1-t)^{\alpha-m-1-\epsilon}.$$

Thus $H^{(m)}(t) \rightarrow +\infty$ and

$$\lim_{z \uparrow R} \frac{\log F^{(m)}(z)}{\log(R-z)} = \alpha - m - 1 \notin \mathbb{Q}. \quad (27.2)$$

The change from H to F only introduces a constant derivative factor and a linear rescaling of distance to the singular point.

27.3 Arithmetic local exponents give the contradiction

Suppose F were D-finite over $\mathbb{Q}(z)$. Its integer coefficients and exponential upper bound make it a G-function: conjugates have the same bound and common denominators are one. Its m th derivative is also a G-function. The arithmetic local-structure theorem of André–Chudnovski–Katz, in the formulation [8, Theorem 6 and Corollary 1, pp. 328–329], says that the minimal operator is regular singular with rational exponents. At the algebraic point R , on the branch approached along $z < R$, its local expression is a finite sum of rational powers times logarithmic powers and holomorphic factors. Combining equal exponents and taking first nonzero Taylor coefficients gives

$$F^{(m)}(z) = c(R-z)^q (\log(R-z))^k (1 + o(1)), \quad q \in \mathbb{Q}, \quad k \geq 0, \quad c \neq 0.$$

On the chosen real interval the leading sign is positive, because $F^{(m)}$ is positive. Its logarithmic limit is therefore $q \in \mathbb{Q}$, contradicting (27.2). Possible operator singularities inside the convergence

disk do not matter: the original power-series solution is analytic through them and determines this branch at R .

The exclusion also holds over $\mathbb{C}(z)$. For a fixed order and degree, the coefficients of a putative differential operator solve homogeneous linear equations with rational entries, since F has rational coefficients. Their row space is finite dimensional and has a finite rational basis. A nonzero complex null vector therefore entails a nonzero rational null vector, reducing to the contradiction above.

This proof uses positivity to extract a real singular logarithmic exponent. It does not invoke a full coefficient equivalent or assert that arbitrary D-finite functions have only rational exponents. The G-function arithmetic hypotheses are essential.

Remark 27.1 (write, 2026-10-05: the criterion in general). *Let $a_n \geq 0$ be integers with $a_n = \Gamma^n n^{-\alpha+o(1)}$, $\Gamma > 1$ and $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. Then $\sum_n a_n z^n$ is not D-finite over $\mathbb{C}(z)$.* This is what the argument above proves; it is stated there for the two sequences, with $m = \lfloor \alpha \rfloor$. In general take any integer $m \geq 0$ with $m > \alpha - 1$. The coefficients of $H^{(m)}$ are $n(n-1) \cdots (n-m+1)a_n R^{n-m}$, eventually positive and equal to $n^{m-\alpha+o(1)}$, and $m-\alpha > -1$. For $0 < \epsilon < m-\alpha+1$, the comparison (27.1) with exponents $m-\alpha \pm \epsilon$ gives the two-sided bound above, so $H^{(m)}(t) \rightarrow +\infty$ (the exponent $\alpha-m-1$ is negative) and the limit (27.2) equals $\alpha-m-1$, which is irrational. The G-function step needs only integer coefficients and an exponential bound, both of which hold, and gives a rational limit, a contradiction; the reduction from $\mathbb{C}(z)$ to $\mathbb{Q}(z)$ is the paragraph above. Lemma 9.2 is the special case $a_n = \Theta(\mu^n n^{-\alpha})$, and Theorem 1.1 with Lemma 9.1 is not needed for the three sequences once (23.1)–(23.3) are known. The bounds $4 < \alpha_P < 5$ and $6 < \alpha_S < 7$ above are rechecked: $\cos(\pi/3) = 1/2 < 9/16 < \cos(\pi/4)$, and $\cos(\pi/5) = (1 + \sqrt{5})/4 < 22/27 < \cos(\pi/6)$, the last because $(44/27)^2 = 1936/729 < 3$.

28 First thresholds without monotonicity

Let $a_n = \Gamma^n n^{-\alpha+o(1)}$ with $\Gamma > 1$ and put $g = \log \Gamma$. The estimate gives eventual positivity and unboundedness, so $N(y) = \min\{n : a_n \geq y\}$ is defined and tends to infinity. First-threshold minimality itself implies

$$a_{N(y)-1} < y \leq a_{N(y)}$$

for large y . Since

$$\log a_n = gn - \alpha \log n + o(\log n),$$

the two adjacent bounds first imply $N(y) \sim (\log y)/g$ and then

$$\log y = gN(y) - \alpha \log N(y) + o(\log N(y)).$$

Substitution of the first-order estimate proves (23.4). In particular no estimate on successive coefficient ratios, derivative of an error term, or eventual monotonicity is assumed. The unknown $o(\log n)$ coefficient remainder is exactly why the inverse remainder is only $o(\log \log y)$, rather than $O(1)$.

[write, 2026-10-05] This is weaker than Lemma 9.5, which gives the same two terms with error $O(1)$ from the Θ bounds, and than Theorem 20.1. All three are instances of `p0:thm:lambert-core` of the transseries volume named in the guide, here at logarithmic precision; no novelty is claimed for the inversion mechanics.

29 A six-face sleeve and the all-index A377921 result

Write $b_n = \text{A377921}(n)$ for the subfamily in which all three white outer vertices are nonisolated. Family inclusion gives $b_n \leq s_n$. The generating-function relation alone only controls a four-term window. The following explicit injection supplies the missing all-index lower bound.

29.1 The source conventions

In the source's definitions [1, Section 2.1], a $(6, 4)$ -dissection has a simple hexagonal outer boundary and all inner faces quadrangular. Its root vertex is white. The outer labels are R, G, B, R, G, B counterclockwise from the root. Every inner edge incident to an outer vertex has that vertex's prescribed label (SL1). At each inner vertex, incident colors form three nonempty clockwise blocks R, G, B (SL2). The size is the number of inner faces. Remark 3 gives the equivalent underlying condition of simplicity and faciality of every four-cycle.

Proposition 29.1 (Canonical sleeve injection). *For every genuine size there is an injection*

$$\Phi : \mathcal{S}_n \hookrightarrow \tilde{\mathcal{S}}_{n+6}$$

on the rooted labeled map classes of the source. In particular,

$$s_{n-6} \leq b_n \leq s_n \quad (n \geq 8). \quad (29.1)$$

Proof. Index the old outer vertices $v_0, \dots, v_5$ counterclockwise, with v_0 the white root. Indices below are modulo six; write $c_i = R, G, B$ according to i modulo three. Thus v_i is white exactly for even i and its prescribed outer color is c_i .

Attach a topological annulus outside the old boundary. Its new outer cycle is $w_0, \dots, w_5$ in the same cyclic order, and its six radial edges are $v_i w_i$. The six new quadrangular faces are

$$v_i v_{i+1} w_{i+1} w_i \quad (0 \leq i < 6).$$

This construction is independent of the geometry of a particular planar drawing. Give w_i the bipartite color opposite to v_i , keep all originally inner edge colors, and color the newly inner edges by

$$\text{col}(v_i v_{i+1}) = c_i, \quad \text{col}(v_i w_i) = c_{i+1}. \quad (29.2)$$

Leave the new outer edges uncolored. Root the result at the unique outer corner of w_5 , which is white. In counterclockwise order from this new root,

$$(w_5, w_0, w_1, w_2, w_3, w_4) \text{ have labels } (R, G, B, R, G, B).$$

Thus the label of w_i is c_{i+1} . Its only inner edge is its radial edge, of exactly that color. Each w_i has degree three, so all six new outer vertices are nonisolated and (SL1) holds.

Originally inner vertices have no change in their rotations or colors. At a former boundary vertex v_i , the clockwise cyclic order is the preceding boundary edge, all old inner edges, the following boundary edge, and the radial edge. If $k_i \geq 0$ old inner edges meet v_i , their colors are all c_i by old (SL1). The new clockwise color list is

$$c_{i-1}, \quad (c_i)^{k_i}, \quad c_i, \quad c_{i+1}.$$

These are exactly three nonempty blocks in cyclic R, G, B order. The following old boundary edge keeps the middle block nonempty even if $k_i = 0$. Thus (SL2) holds also at formerly isolated outer vertices.

The new bipartition is consistent on boundary edges and radials, the outer cycle is simple, and all old inner faces are unchanged. There are no new multiple edges. One can also check irreducibility directly. A four-cycle containing no new vertex is an old facial cycle. A four-cycle using both old and new vertices crosses the radial cut an even number of times. It cannot use four radials, since each new vertex has only one radial edge, so it uses exactly two. The two remaining edges must be one outer-ring edge and one old edge. The radials consequently have consecutive indices, and the old edge is the corresponding old boundary edge by simplicity. The cycle is one of the six annular faces. A four-cycle entirely among new vertices is impossible in their chordless hexagon. Hence every four-cycle is facial.

Exactly six vertices, twelve edges and six inner faces were added. The size increase is precisely six. There are three new vertices of each bipartite color, so the refined size also changes by $(a, b) \mapsto (a + 3, b + 3)$.

Finally the construction has an intrinsic rooted inverse on its image. Each outer vertex of an output has a unique inner neighbor, recovering the six old boundary vertices. The new boundary vertex immediately counterclockwise after the new root is w_0 ; its unique inner neighbor is the old root v_0 . Delete the six new outer vertices and all incident edges, expose the recovered old hexagon, erase the colors on its boundary edges, and root at its outer corner at v_0 . Every old inner edge color is retained. This recovers the input exactly. The procedure respects orientation-preserving rooted isomorphisms, so symmetry creates no multiplicity. This proves injectivity and the sandwich. $\square$

Corollary 29.2 (A377921 at every large index). *The sequence $b_n = \text{A377921}(n)$ satisfies*

$$b_n = (16/3)^n n^{-\alpha_S + o(1)}. \quad (29.3)$$

Its generating function is not D -finite over $\mathbb{C}(t)$, and

$$N_b(y) = \frac{\log y}{\log(16/3)} + \frac{\alpha_S}{\log(16/3)} \log \log y + o(\log \log y).$$

Proof. The fixed six-unit shift obeys

$$\log s_{n-6} = n \log \Gamma_S - \alpha_S \log n + o(\log n).$$

Taking logarithms in (29.1) proves the all-index estimate. The positive-series G -function and threshold arguments in Sections 27–28 apply directly. $\square$

[**write, 2026-10-05**] Part II proves the equivalent $b_n = \tilde{s}_n \sim (16/19)^3 \kappa_S \mu_S^n n^{-\alpha_S}$, (11.4), by a dominated signed convolution from the equivalent for s_n ; the corollary is a second route on the logarithmic scale that needs no amplitude. The two are consistent: at $n = 30$, $b_{30}/s_{30} = 0.5429 \dots$ against the limit $(16/19)^3 = 0.5971 \dots$, and $s_{24} \leq b_{30} \leq s_{30}$.

Remark 29.3 (write, 2026-10-05: a Θ bound for A377921 from Part I). $b_n = \Theta(\mu_S^n n^{-\alpha_S})$. Indeed $b_n \leq s_n$, and $b_n \geq s_{n-6}$ for $n \geq 8$ by Proposition 29.1. Theorem 1.1 gives positive c, C with $c\mu_S^n n^{-\alpha_S} \leq s_n \leq C\mu_S^n n^{-\alpha_S}$ for large n , hence $b_n \geq c\mu_S^{-6} \mu_S^n (n-6)^{-\alpha_S} \geq c\mu_S^{-6} \mu_S^n n^{-\alpha_S}$. So the coefficientwise Θ estimate that Part I says does not follow from its argument (after Corollary 9.4) follows from it together with the sleeve, without Part II.

29.2 The independent positive-convolution consequences

For comparison, set $A(t) = \sum_{n \geq 2} s_n t^n$ and $B(t) = \sum_{n \geq 2} b_n t^n$. The source's exact relation [1, Remark 20] is

$$A(t) = (1+t)^3(1+B(t)) - 1 - 3t. \quad (29.4)$$

For $n \geq 4$ this yields

$$s_n = b_n + 3b_{n-1} + 3b_{n-2} + b_{n-3}. \quad (29.5)$$

Nonnegativity gives

$$s_n/8 \leq \max_{0 \leq j \leq 3} b_{n-j} \leq \max_{0 \leq j \leq 3} s_{n-j}.$$

Thus even without the sleeve one obtains the logarithmic estimate for the four-term maximum, $\limsup b_n^{1/n} = \Gamma_S$, and $N_s(y) \leq N_b(y) \leq N_s(8y)$ for large y . Also D-finiteness of B would force D-finiteness of A by (29.4), giving a short independent transfer of non-D-finiteness.

The distinction is important: a short positive convolution can hide small coefficients in a parity class. Dividing by $(1+t)^3$ introduces alternating coefficients. Neither maneuver proves an all-index lower bound; that bound comes from the explicitly rooted sleeve injection. The injection does not supply a limiting ratio b_n/s_n , a leading amplitude, or a full coefficient equivalent.

Remark 29.4 (write, 2026-10-05: the same identity as Part I). (29.4) is (9.4): with $A = S$, $B = T$ and $t = z$, $(1+t)^3(1+B) - 1 - 3t = (1+t)^3B + (1+3t+3t^2+t^3) - 1 - 3t = (1+t)^3B + 3t^2 + t^3$. Report 112's caution is borne out by Part II: its proof of (11.4) divides by $(1+z)^3$, as here, and controls the alternating coefficients only by dominated convergence against the full equivalent for s_n ; a Θ or logarithmic bound would not suffice for that step. The limiting ratio is $(16/19)^3$; Part II gives no amplitude digits.

30 Why a full equivalent is still outside the result

The logarithmic theorem loses arbitrarily small powers at its annular crossings. Its proof consequently identifies the exponent but cannot recover a stable amplitude. A sufficient stronger result would be a fixed-endpoint killed local limit of the form

$$K_n((x, c), (y, d)) \sim \kappa V_c(x) V_d^*(y) n^{-1-\pi/\theta},$$

with positive forward and dual killed harmonic functions and the correct lattice normalization. This is a different analytic task.

A scoped review of primary results found no theorem immediately covering these exact two-dimensional lattice kernels with their finite phases and unbounded edge-valued increments:

- Denisov–Wachtel's *Random walks in cones revisited* [2] assumes independent identically distributed increments. Its cone harmonic and survival theory cannot simply be imported as a finite-phase fixed-endpoint result.
- Denisov–Zhang's *Markov chains in the domain of attraction of Brownian motion in cones* [4] gives harmonic functions, survival equivalents and a conditioned global endpoint limit under conditional moment assumptions. It does not state the required fixed-endpoint killed local limit, and its conditional covariance assumptions fail for the exact kernels here.
- Grama–Lauvergnat–Le Page [5] proves a conditioned local limit for a scalar additive functional on a finite Markov chain, killed at a half-line. Its Theorem 2.5 is not a theorem for a two-dimensional wedge or these unbounded edge increments.
- The March 2026 paper of Pham–Peigné–Son [6] treats nonlattice multidimensional random walks in cones; its setting is not the present lattice internal-phase process.

This is a statement about the inspected hypotheses, not a claim that no other applicable result exists.

The covariance obstruction in the second item is exact. After the natural Poisson corrector, the phase-conditional covariance matrices are

$$\begin{aligned} V_0^P &= \begin{pmatrix} 3/5 & -9/20 \\ -9/20 & 1 \end{pmatrix}, & V_1^P &= \begin{pmatrix} 1 & -9/20 \\ -9/20 & 3/5 \end{pmatrix}, \\ V_0^S &= \begin{pmatrix} 3/2 & -22/21 \\ -22/21 & 15/14 \end{pmatrix}, & V_1^S &= \begin{pmatrix} 15/14 & -22/21 \\ -22/21 & 3/2 \end{pmatrix}. \end{aligned}$$

Their stationary averages are Σ_P, Σ_S , but the two matrices in each pair are unequal. No common invertible whitening can make both identity matrices. Their difference persists arbitrarily deep in the cone because the transition law depends only on phase. A martingale correction also shifts the boundary by a bounded phase-dependent amount; it does not turn the exact killed process into one on a single unchanged cone. The logarithmic proof requires only the averaged FCLT covariance and handles the original boundary throughout.

Regeneration is not an automatic remedy. A return block must retain its duration and its within-block coordinate minima. Omitting the duration changes the counting parameter; omitting the minima admits paths that left the quadrant inside a block. A full-equivalent proof would need to handle these marks or develop a phase-sensitive killed harmonic and local-limit theory directly.

[write, 2026-10-05] For the two models of this report the full equivalent is no longer outside the report. Part II proves it by regeneration that keeps exactly the marks named here: the cycle durations, by a two-sided random clock (Theorem 16.1), and the within-cycle minima, through the valid complete-cycle kernel D , squeezed as $P_b \leq D \leq P_0$ between an iid cone kernel and its translate ((13.5)), which the one-unit buffer of Lemma 4.1 makes possible. It proves the killed local limit from the free local theorem and an exit decomposition (Lemma 17.1), and the fixed-endpoint form asked for at the start of this section is Theorem 18.1, $q_n(a, b) \sim \mathcal{B}_\Sigma A(a)A^*(b)n^{-p-1}$ with positive harmonic A, A^* . Its proof uses Denisov–Wachtel [2] for the iid cycle walk, not [4], [5] or [6]. The review above remains an accurate account of the papers it inspected; its sentence that no theorem covers these kernels was true of the literature it checked, not of this report. In the generality of Theorem 24.1 the amplitude question is open (Question Q1). The phase-conditional matrices displayed above are (3.5) and (3.6) divided by 4, and Part I draws the same conclusion from them (after (3.7)). Report 112’s bibliography cites [6] from its abstract and paper record only.

31 Further research and unresolved precision

The following are concrete next problems, not conclusions of the present proof.

1. **Leading amplitudes and a killed local limit.** Construct positive phase-indexed forward and dual harmonic functions for the exact killed chains, and establish a fixed-endpoint equivalent with its lattice normalization. This would strengthen the logarithmic theorem rather than merely repeat the unrestricted local limit.

[write, 2026-10-05] Solved for the two models by Part II: Theorem 16.1 (positive harmonic functions A, A^* at every relevant endpoint), Lemma 17.1 and Theorem 18.1 (the fixed-endpoint equivalent, with the lattice factor two). Open in the generality of Theorem 24.1: Question Q1.

2. **A second phase correction.** A possible route is to solve a finite-state Poisson equation for the phase discrepancy $V(c) - \Sigma$ and couple the solution to the Hessian of the Brownian wedge harmonic function. This may cancel the persistent second-order phase oscillation. Boundary

control, positivity, convergence and the fixed-time local equivalent would still require separate proofs.

[write, 2026-10-05] Not pursued in Parts I–II, which reach the amplitudes through regeneration instead. The proposed route stays open as stated, with what is missing listed by Report 112 itself (boundary control, positivity, convergence, the fixed-time equivalent); it would matter for chains without a cycle buffer.

3. **Correction terms.** Only after a leading equivalent is established should one determine which further powers or logarithms occur, and whether the finite-phase boundary contributes additional terms. No correction hierarchy or transseries is inferred from $n^{o(1)}$ here.

[write, 2026-10-05] Still open. The leading equivalents now exist (Theorem 11.1); no rate and no correction term is proved in any Part.

4. **The restricted-to-unrestricted ratio.** The sleeve proves the exponent for b_n , but not a limit for b_n/s_n . Determining that ratio, or its connection to possible leading amplitudes through the exact generating-function identity, requires stronger coefficient control that rules out alternating cancellation.

[write, 2026-10-05] Solved: the limit is $b_n/s_n \rightarrow (16/19)^3$, by (11.3)–(11.4) and the dominated signed convolution of Part II, the kind of coefficient control asked for here. With the sleeve, Part I alone already gives $b_n = \Theta(\mu_S^n n^{-\alpha_S})$ (Remark 29.3).

5. **Sharper first thresholds.** An $O(1)$ remainder in the inverse would require at least bounded logarithmic coefficient error. Further constant-order inverse terms would also need a precise amplitude and appropriate control of the discrete first crossing. These data are currently absent.

[write, 2026-10-05] The $O(1)$ remainder is Lemma 9.5, from the Θ bounds. The constant-order term is Theorem 20.1: Lambert- W_{-1} centering with the next constant $-(\alpha \log \log \mu + \log \gamma)/\log \mu$ and a vanishing-width two-ceiling bracket. Still open: an exact rounding rule $N = \lceil r \rceil$, which needs a quantitative remainder in the forward asymptotic, and any numerical value of the amplitudes that enter it.

31.1 Further questions recorded at the write

Under Vladimir’s standing rule of 4 October 2026, every claim of Report 112 that is not proved in this report is recorded here or above with its source. Report 112 makes no claim that the write found to be wrong.

- Q1. *An amplitude form of Theorem 24.1.* Under (H1)–(H5), possibly with a further hypothesis, is $K_n((0, a), (0, b)) \sim \kappa V_a(0) V_b^*(0) n^{-1-\nu}$ with positive phase-indexed killed harmonic functions? Report 112, Sections 30–31, poses it; Part II answers it for the two models, using a fixed cycle buffer (Lemma 4.1) that a general chain need not have. Missing: a harmonic comparison without a buffer, or a phase-sensitive killed local limit theory (Report 112’s own suggestions); the crossing argument of Section 24.4 loses $q^{\pm\eta}$ per annulus and cannot give a constant.
- Q2. *A Θ form.* Do (H1)–(H5) give $K_n((0, a), (0, b)) = \Theta(n^{-1-\nu})$? Part I proves this for the two models by regeneration and the buffer. The proof of Theorem 24.1 gives no rate for its $o(1)$ (Report 112 says so).

- Q3. *Endpoints without seeds.* Report 112 notes that a centred chain can have a boundary phase from which no legal interior excursion exists, and makes no claim there. What is the exponent, if any, at such endpoints?
- Q4. *Steps not independently rechecked at the write.* The uniform transfer from Brownian motion near the sides of the wedge (Section 24.4), the bridge tightness in the proof of Lemma 24.2, and the page locators of [3, Theorem 2.1(ii)] and [8, Theorem 6, Corollary 1] are given by Report 112 with the arguments printed above; the intake read them as sound but did not verify the cited pages. The check that Report 111’s chain satisfies (H1)–(H5) (Section 22.5) is printed in that report, not here.
- Q5. *Rates, corrections and rounding* (Report 112’s items 3 and 5): open, as in the notes above.

32 Reproducibility and interpretation of the checks

The accompanying reader package contains this editable L^AT_EX source, the PDF, exact symbolic and finite-enumeration checks, dependency instructions, a source ledger and a strict checksum manifest. It contains no third-party source PDFs. The package README gives the build and replay commands.

The computational checks reconstruct both rational kernels and their normalizations; verify stationary centering, correctors, effective and phase-conditional covariances; check exponential-moment witnesses, support edges, endpoint factors and shifts; verify finite sleeve topology and rooted recovery cases; and reproduce the published finite prefixes from the original walk descriptions. They run with ordinary Python and with optimization enabled, without relying on removable `assert` statements. Deliberately damaged fixtures are required to fail, and a fresh extracted archive is replayed. The PDF build fixes its date and metadata inputs.

For corner paths, finite enumeration prunes a state at time k when $y > N - k$: only SE lowers y , by one, so the state cannot contribute to any horizontal-axis endpoint by time N . This proves the requested p_n prefix through N , but it does *not* certify an auxiliary $(1, 1)$ endpoint count at time N with that same pruning. The checker must either enlarge the height budget for such auxiliary endpoints or omit the truncated boundary value.

For Schnyder paths, each SE-count layer is closed under face steps. Every face strictly decreases x , so processing x in descending order is acyclic and exact. Binomial multiplicities and the SE time parameter are retained. The four exceptional initial s_n values are entered explicitly; the positive relation is used only from $n = 4$. The formal auxiliary initialization $s'_0 = 1$ is not promoted to a combinatorial size-zero count.

Finite checks certify the listed algebraic identities and enumerated ranges, not the asymptotic theorem. The all-index conclusions depend on the proofs in this report and the cited source bijections. Neither matching a long prefix nor passing a mutation suite proves a cone asymptotic. Conversely the proof’s analytic steps do not depend on numerical fitting or extrapolation from those prefixes.

[**write, 2026-10-05**] In this repository the package’s PDF, README, source and checksum inventory (`manifest.json`) are not shipped; its programs are shipped as `code/47-logcone-*.py`, its fixtures and recorded outputs as `data/47-logcone-*`, and its two check notes at the report root. The checker imports `verify_sleeve` and reads `fixtures.json` beside itself, and the integrity, seal, build and replay programs need the delivered layout, the manifest or the original archive, so every rerun is made on a copy; the report README gives the routes. Rerun on 5 October 2026 on such a copy, `checks/validate.py` reproduced `data/47-logcone-checks-results-ordinary.log` and `.json` modulo line endings. Besides

the moment $189/13$ of (26.7), the checks certify the corner quotient-displacement moment $(2/3)R + (4/27)R/(1 - R/3)^2 = 524/405$ at $R = e^t = 6/5$, not stated in the text. The fixed-endpoint counts e_L are enumerated with the enlarged height budget described in the paragraph on corner paths above.

A Exact computational checks

The scripts require Python 3 and SymPy. All paths in the reproducibility package are relative to the package root. Running `python3 scripts/run_all.py` regenerates the verification outputs, and running `bash build.sh` recompiles the article.

[**write, 2026-10-02**] These are delivery paths. In this repository the scripts are shipped as `code/45-cone-*.py`, their outputs and the OEIS excerpts as `data/45-cone-*`, and `build.sh` as `code/45-cone-build.sh`, which builds a delivery-layout `article.tex` (source 45 alone). The scripts resolve `scripts/`, `results/` and `sources/` relative to their parent directory, so they must be run on a scratch copy in the delivered layout; the report README gives the recipe.

A.1 Symbolic identities

The exact-arithmetic checks differentiate the two rational kernel matrices and their return generating functions. They verify stochasticity, the conditional drifts, both forward and reverse correctors, each conditional covariance matrix, the zero cycle means, both parity cycle covariances, the mean and variance of the cycle duration, the covariance angles, and the S aggregate resolvent identity. No floating-point approximation is used for these assertions. Decimal exponents in the results table are evaluated only after the exact formulas have been established.

A.2 Finite cycle geometry

The supplied independent P audit exhaustively tests the one-unit minimum bound and its reversed form for 25,900 nontrivial cycles with bounded face-step parameters and at most three middle parity-preserving steps. This finite test is useful for detecting sign errors. The proof for unbounded cycles is Lemma 4.1, not an extrapolation from the test.

A.3 Integer enumeration

For P , an elementary dynamic program counts admissible physical walks. When counting walks with total length N , a state at time t with ordinate greater than $N - t$ can be discarded: only a southeast step can decrease the ordinate, and it does so by one. Nonnegative horizontal coordinates bound possible face-step horizontal decrements. These restrictions make every counting sum finite without approximation.

For S , the package includes the weighted southeast/face recurrence and a separately implemented direct enumeration check. The weights $\binom{\ell+r}{r}$ are exact integers. The relation (2.8) converts the restricted counts s'_n to s_n . Initial coefficients provide explicit indexing checks:

n	p_n	s_n
0	0	0
1	0	0
2	0	3
3	1	2
4	0	3
5	3	6
6	4	14
7	15	36
8	39	102
9	122	306
10	375	972
11	1212	3216
12	3980	11050
13	13413	39138
14	45966	142446
15	160295	530656

The longer recorded Schnyder recurrence output reaches $n = 30$ and agrees with the supplied OEIS export. Comparisons are exact coefficient comparisons; no numerical fit is used to infer either exponent.

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